BioHackrXiv

Predicting microbial energy metabolism from genomes through knowledge graphs and machine learning

Shuichi Kawashima [1], **Danil Ezhov** [2], **Akira R Kinjo** [3], **Julia Koblitz** [4], **Takeru Nakazato** [5], **Yoko Okabbepu** [6], **and Risa Otsuka** [5]

1 Database Division for Life Science, BioData Science Initiative, National Institute of Genetics, Research Organization of Information and Systems, Chiba, Japan **2** ITMO University, Russia **3** Anima Machina G.K., Osaka, Japan **4** OSIRIS Solutions GmbH, Germany **5** Biological Resource Center, National Institute of Technology and Evaluation, Japan **6** OKBP Inc., Japan

BioHackathon series:
DBCLS BioHackathon 2026
Matsuyama, Japan, 2026
microbial-metabolic-traits

Submitted: 23 Oct 2026

License:
Authors retain copyright and release the work under a Creative Commons Attribution 4.0 International License (CC-BY).

Published by BioHackrXiv.org

Abstract

The increasing availability of microbial genomes creates opportunities to predict physiological traits and prioritize cultivation experiments. During DBCLS BioHackathon 2026, we connected genome-derived protein annotations to electron donor and acceptor information, initially focusing on autotrophic microorganisms. KofamScan annotations for 23,434 RefSeq genomes and KEGG Orthology (KO) definitions were represented in RDF and made accessible through TogoMCP. Literature-assisted phenotype collection and an interactive viewer linked reported physiology to genomic evidence. Within a focused collection of 586 genomes, 358 labeled genomes supported evaluation of 14 metabolic categories. A KO-based classifier achieved macro area under the precision-recall curve (AUPRC) of 0.612 when entire genera were held out, compared with 0.472 for a taxonomic baseline. Combining KO and protein language model representations improved internal performance to 0.625, although the contribution beyond ensembling KO-based learners was not statistically resolved. This improvement did not transfer to external evaluation. On 562 genomes from unseen genera, the KO model achieved 0.555 versus 0.380 for the taxonomic baseline. Completeness scores for 71 KEGG modules provided complementary pathway annotations. Performance varied substantially among traits, and disagreement between phenotype compilations identified priorities for further curation. These hackathon outcomes establish a shared basis for evidence retrieval, comparative visualization, and genome-guided screening, while experimental validation remains necessary before predictions can guide cultivation.

Introduction

Metagenomic sequencing and genome reconstruction have greatly expanded access to microbial diversity, including lineages without cultured representatives. Large collections of metagenome-assembled genomes (MAGs) provide opportunities to investigate functions that would be difficult to discover through cultivation alone (Nayfach et al., 2021). However, the increasing availability of sequences has not been matched by physiological characterization. Methods that connect genomes to experimentally relevant traits could help prioritize organisms for screening and identify candidates with useful biochemical capabilities.

Carbon fixation provides one example of this approach. AutoFixMark uses curated combinations of KEGG Orthology (KO) markers to infer CO2 fixation pathways from genome annotations (Kawashima et al., 2026). Such resources make biological knowledge usable for interpretable screening. The ability to assimilate inorganic carbon, however, does not identify how an organism acquires the energy and reducing power needed for growth. Carbon assimilation and energy acquisition therefore require complementary descriptions.

Electron donors and acceptors are central to this second question. Their utilization describes the redox transformations through which microorganisms can conserve energy and can inform the selection of candidate medium components and cultivation atmospheres. We initially focused on autotrophic microorganisms, particularly chemolithoautotrophs, because existing carbon fixation resources offer a practical starting point and because considering carbon supply separately from electron donation helps formulate explicit cultivation hypotheses. This focus does not imply that all autotrophs use exclusively inorganic electron donors or that carbon fixation genes alone establish autotrophic growth. Facultative lifestyles and condition-dependent substrate use remain relevant.

Several tools already predict microbial traits from genomes. Traitar learns protein-family signatures for 67 phenotypes from labeled isolates (Weimann et al., 2016), ProTraits combines text mining with comparative genomics for several hundred traits (Brbić et al., 2016), and FAPROTAX, DRAM and METABOLIC assign functions from taxonomy or curated annotation rules (Louca et al., 2016; Shaffer et al., 2020; Zhou et al., 2022). Other predictors estimate optimal growth temperature, maximal growth rate or oxygen use from genome composition (Flamholz et al., 2024; Sauer & Wang, 2019; Weissman et al., 2021), and metabolic reconstructions predict substrate use (Zimmermann et al., 2021). Classifiers trained on curated strain data from BacDive now add predicted traits to that database (Koblitz et al., 2025), and Microbe Decoder offers taxonomy-based, metabolic-network and machine-learning predictors evaluated with higher taxonomic ranks held out (Hackmann et al., 2026). These resources cover respiration types and fermentation well. Electron donors used by lithotrophs, such as hydrogen, reduced sulfur, ferrous iron, ammonia, nitrite and carbon monoxide, are rarely predicted from gene content and tested outside the lineages used for training.

At BioHackathon 2026, we combined data integration, literature-assisted curation, visualization, and predictive modeling around this question. The project addressed two related needs: enabling an AI assistant to retrieve and explain evidence about a named organism, and evaluating whether genome-derived features predict traits beyond labeled training lineages. The former can draw on published physiology, whereas the latter must be assessed under controlled conditions relevant to uncharacterized organisms. We report the resources developed during the hackathon, trait predictors evaluated on lineages and species absent from training and on metagenome-assembled genomes, their comparison with existing tools, and the implications for genome-guided screening.

Methods

Genomic and physiological evidence

The project used a broad resource of KofamScan annotations for 23,434 RefSeq reference genomes and a focused collection of 586 autotrophic microbial genomes. Assembly accessions provided the connection between genomic annotations and physiological records. The focused benchmark contained 358 genomes with electron donor and acceptor labels; the remaining 228 genomes had genomic annotations but lacked the benchmark phenotype labels.

KofamScan assigns KO identifiers to proteins using profile hidden Markov models and profile-specific score thresholds (Aramaki et al., 2020). We retained protein-to-KO assignments together with scores, thresholds, E-values, and significance flags. These records preserve the supporting protein evidence behind a genome-level KO profile. They also permit analyses to select significant assignments without losing the underlying scoring information.

The prediction workflow accepts an NCBI assembly accession or a nucleotide FASTA file as input. It uses proteins from the NCBI annotation or, for unannotated sequences, predicts them with Pyrodigal (Hyatt et al., 2010; Larralde, 2022). It then assigns KOs with the KofamScan rules (Aramaki et al., 2020), searching each protein against the KOfam release of 25 August 2026 with pyhmmer (Larralde & Zeller, 2023). A KO is assigned to a protein

when its full-length or domain score, depending on the profile, reaches the profile's threshold. A genome is assigned a KO when at least one of its proteins is assigned that KO. The same workflow is used to predict donors and acceptors and pathways, and to compute module completeness, for any assembled genome, including MAGs (Fig. 1b).

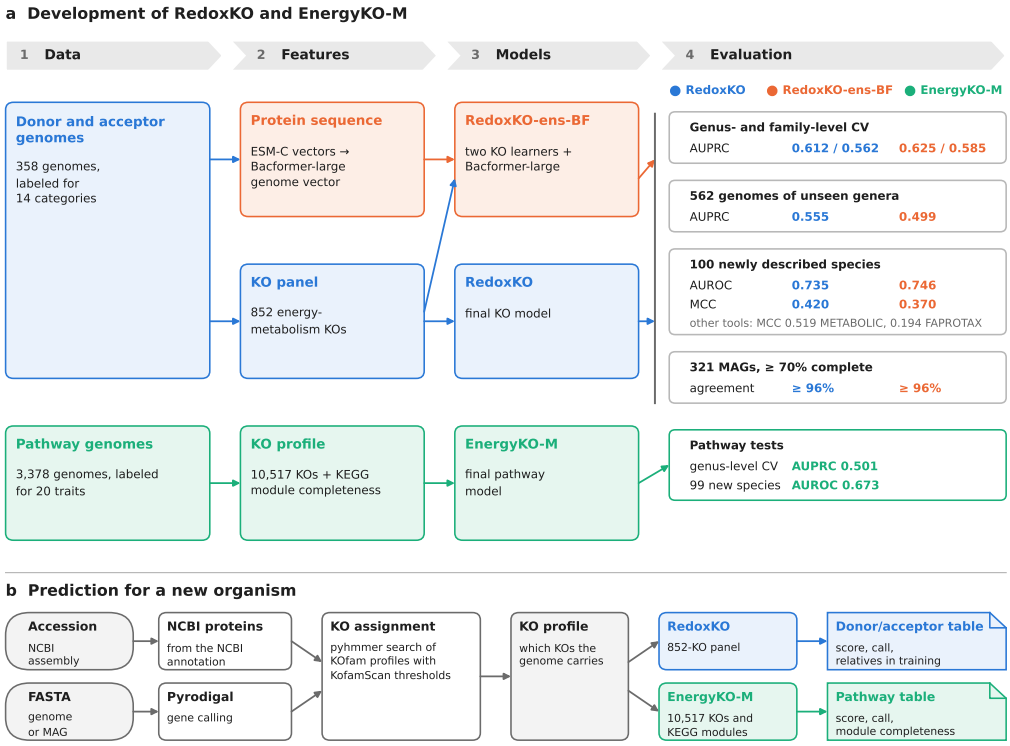


Figure 1: Study overview. (a) Model development. Colors mark the donor and acceptor task (blue), the pathway task (green) and the model with the Bacformer genome vector (orange). (b) Applying the final models to a new organism.

RDF representation and access through TogoMCP

We converted genome-protein-KO associations to Resource Description Framework (RDF), representing assemblies, proteins, and KOs with identifiers.org URIs. The graph retains individual protein assignments as well as assignment-level attributes. This allows retrieval of either a compact KO set for an assembly or the proteins and scores supporting a specific annotated function.

The KofamScan definition file was also converted to RDF. KO descriptions were represented with skos:definition, while profile metadata included score thresholds and score types. Because definitions and assignments use shared KO identifiers, queries can join the observed genomic evidence to a description of its function. An assistant can consequently request relevant subsets of annotations and their definitions instead of receiving an entire proteome annotation table in its prompt.

The RDF data were loaded into QLever and made queryable through SPARQL. Registering the dataset with TogoMCP enabled AI assistants to retrieve these annotations alongside other resources available through RDF Portal (Kinjo et al., 2026), (Kawashima et al., 2018). The implemented integration supplies evidence for questions about the donors and acceptors used by a specified organism. It exposes the annotation resource; deployment of the trained trait classifier as a dedicated MCP prediction tool is a separate step.

Literature-assisted phenotype collection

Workflows assisted by large language models (LLMs) extracted electron donor and acceptor information from research articles and microbial databases, including BacDive (Schober et al., 2025). One of these workflows used FEDOT.MAS (Gilemkhanov et al., 2026), which generates multi-agent LLM pipelines from a task description. A meta-agent (DeepSeek V4 Pro) generated the pipelines, and executor agents (Gemini 3.7 Flash) extracted donors and acceptors for 421 organisms from open-access texts (PubMed, PubMed Central, Unpaywall and Europe PMC), returning each value with a quoted string. We merged complementary searches and extraction efforts, keeping supporting text, evidence grades and conflicting claims. Database fields on nutrition, metabolite utilization and oxygen tolerance supplied further context. These fields need biological interpretation, because oxygen tolerance, for example, differs from evidence of respiratory oxygen use.

The merged collection used four evidence grades. Grade A marked statements of experimental evidence found by at least two extraction efforts, and grade B marked such statements found by one effort. Grade C covered genomic inference, hedged or review-only statements, weak evidence and conflicting claims. Grade D covered unsupported, poorly documented or disputed records. Agreement between extraction efforts does not establish independent experimental replication, and the grades do not imply that every record received expert review.

A snapshot of 16 September 2026 contained 1,817 evidence records, of which 702 were grade A, 921 grade B, 136 grade C and 58 grade D. These are statements, not distinct organisms or independent training examples. The benchmark labels comprised 14 categories, seven for donors and seven for acceptors. Some categories combined related physiology, such as methanogenesis and acetogenesis, and residual categories grouped other donors or acceptors. The evidence collection and the benchmark label matrix therefore represent different levels of aggregation.

In October 2026 we audited the merged collection. Deterministic tools checked that each cited source existed and contained its quote, and a FEDOT.MAS pipeline (Gemini 3.8 Flash as executor) judged the remaining flagged items. We applied a correction only when two independent runs called an item an error and the corrected quote appeared exactly in its source. Most errors concerned the supporting evidence, and fewer the value itself (Supplementary Table S3). After the audit, 2.1% of sampled values had a major error, a wrong value, organism or role. This rate is similar to that of human and LLM-assisted data extraction in systematic reviews (Gartlehner et al., 2025).

Marker references and comparative visualization

We assembled reference tables linking donor and acceptor transformations to relevant genes, enzymes, and KO identifiers. A matching tool compared these markers with genome annotations, and a web viewer displayed genomic markers alongside reported substrates and evidence grades. The viewer was developed for the broad RefSeq resource and the focused autotroph collection.

Additional comparative displays used Jaccard similarity between KO sets to organize genomes by their functional profiles. Such a representation supports exploration of shared annotated capabilities; it should not be interpreted as a sequence-based phylogeny. Together, the interfaces help identify organisms for which literature-derived physiology and expected genomic markers agree, and cases requiring closer examination. Some phenotype entries and associated descriptive annotations remained preliminary at the hackathon cutoff.

Donor and acceptor classifiers

Training labels came from the phenotype collection. Each of the 358 labeled genomes has a yes/no label for 14 categories. Seven are electron donors (sulfur compounds, ammonia, nitrite, H_2, CO, Fe(II) and other donors). The other seven are electron acceptors (O_2, nitrate by

denitrification or DNRA, sulfate, sulfur compounds, CO_2 by methanogenesis or acetogenesis, Fe(III) and other acceptors). Other donors and other acceptors collect all compounds outside the named categories. A genome is positive for a category when the collection records at least one donor or acceptor of that category for the organism, and negative otherwise, so a trait that was never reported counts as absent. The 228 genomes of the focused collection without any record are unlabeled. As a second, independent source of labels we used a published trait compilation (Madin et al., 2020), which merges phenotype data for bacterial and archaeal species from many published sources and which we call the larger dataset. The category labels were assigned by the curators of the collection in the shared genome table of 17 September 2026. They do not record the evidence grade of the underlying statement, so a positive may rest on genome-based inference as well as on a growth experiment.

We represented each genome by the presence or absence of 852 KOs. This panel comprises every KO under KEGG BRITE Energy metabolism (09102) and 84 curated marker KOs, that is, genes chosen from the literature as known markers of each category. We fixed this panel before fitting any model. We did not use labels from the larger dataset for training, for three reasons. First, we used part of it as our external test set, which must stay separate from training. Second, it covers only 9 of our 14 categories. Third, its labels for lithotrophic donors are mostly predicted by FAPROTAX (Louca et al., 2016) from taxonomy, whereas ours come from the literature. To check this decision, we added its genomes, with their labels, to training (Supplementary Table S4), which lowered the ranking of nitrate tests on new species (Discussion).

For each category, we trained a random forest (Breiman, 2001) of 500 trees, averaged over three random seeds, and chose its minimum leaf size by inner cross-validation grouped by taxonomy. We refer to this classifier as RedoxKO. We also trained L2-regularized logistic regression and gradient-boosted trees (XGBoost) (Chen & Guestrin, 2016) on the same KO features. RedoxKO-LX (L for logistic regression, X for XGBoost) used the better of the two for each category, chosen by inner cross-validation, and RedoxKO-ens averaged their probabilities. Both are separate from the random forest RedoxKO.

The categories are imbalanced (Supplementary Table S1). RedoxKO therefore weighted the classes by inverse frequency within each bootstrap sample of the random forest. We also tried undersampling of negatives, oversampling of positives and no weighting, and none performed significantly better (Supplementary Table S8).

We chose decision thresholds that maximized the Matthews correlation coefficient (MCC) (Chicco & Jurman, 2020; Matthews, 1975) on inner out-of-fold predictions, never on the test fold, and calibrated the final scores with Platt scaling (Platt, 1999). A category received a yes/no call only if it met four conditions. First, in family-level cross-validation, the lower bound of the 95% confidence interval of its MCC had to exceed 0.2. Second, in the same cross-validation the model had to outperform the taxonomic baseline, with a Benjamini-Hochberg (BH) adjusted q below 0.05. Third and fourth, where the external genomes have labels for the category, its AUPRC on them had to exceed the share of positive genomes, and the MCC of its calls on these genomes had to exceed 0.2 as well. We added the fourth condition on 22 October 2026, after all other results were known (Supplementary Note 1). Supplementary Table S6 gives these inputs for each category. The external genomes therefore took part in deciding which categories get calls. Ammonia oxidation has no external labels and no tests on new species, so its calls were checked only in cross-validation. For categories that fail any of the conditions, the workflow returns a score but no call.

Sequence representations and fusion

We compared three ways to represent a genome by its protein sequences. As a simple baseline used in earlier work (Hwang et al., 2024), we summarized protein embeddings from ESM-2 (35M parameters) (Lin et al., 2023) and, in a separate model, from ESM-C (300M) (EvolutionaryScale, 2024). The summary was the average or the maximum over all proteins of

a genome, chosen by inner cross-validation. We also weighted the ESM-2 embedding of each protein with attention-based multiple-instance learning (Ilse et al., 2018), and used the genome vectors of Bacformer-base and Bacformer-large (Wiatrak et al., 2025). Bacformer reads ESM-C protein embeddings in genomic order and, like other genome language models, represents each protein in its genomic context (Hwang et al., 2024; Nguyen et al., 2024). To predict traits from the resulting genome vector, we compared logistic regression with tree-based, tabular and neural learners and with fine-tuning of Bacformer-large (Supplementary Table S7). None did clearly better, so we kept logistic regression.

Combined models averaged the probabilities of a KO model and a sequence model with equal weights. The strongest fusion, RedoxKO-ens-BF (BF for Bacformer), combines RedoxKO-ens with the Bacformer-large classifier, and RedoxKO-ens-ESM combines RedoxKO-ens with the ESM-2 model in the same way. Taxonomy, organism names and literature were never inputs to the supervised models.

Pathway classifiers and module completeness

For pathway traits, we used every species in the larger dataset (Madin et al., 2020) that has pathway labels and an annotated RefSeq genome, one genome per species (Supplementary Table S1). The phenotype collection covers donors and acceptors, and only part of its genomes have pathway labels. A pathway model trained on these genomes alone could cover only a few of the 20 traits and was less accurate (Supplementary Table S13). We therefore used the larger dataset, although many of its labels, including most lithotrophic ones, are predicted by FAPROTAX (Louca et al., 2016) from taxonomy. We merged its pathway terms into traits and kept the 20 with at least 25 positive genomes. For these genomes, and for the LLM prompts, we used the KofamScan annotations of the shared resource, which agreed with our own annotation in 99.95% of genome-KO pairs on 306 genomes annotated both ways. Before fitting any model, we split the genomes into five folds by genus and, separately, by family.

We then compared three feature sets, the 852-KO panel, the full profile of 10,517 KOs (EnergyKO), and the full profile plus the completeness of KEGG energy modules (EnergyKO-M, M for modules). We also tested a multi-label neural network, random forests (Breiman, 2001), extra-trees (Geurts et al., 2006) and EnergyKO fused with Bacformer-large (EnergyKO-BF). Each pathway model chose for each trait between L2-regularized logistic regression and XGBoost by inner cross-validation, as RedoxKO-LX does. We used the same baselines and metrics as for donors and acceptors, and the first two conditions of the release rule, because the pathway traits have no external test. Following the registered rule, the final pathway model was the registered model with the highest macro AUPRC in genus-level cross-validation, EnergyKO-M. Extra-trees were added after this choice.

Separately, we scored the completeness of 71 KEGG modules covering energy metabolism, carbon fixation, and methane, nitrogen and sulfur metabolism (Kanehisa et al., 2025). Each score is the share of a module's steps covered by the genome's KOs. It is read directly from the annotation.

Cross-validation and evaluation metrics

We ran cross-validation at two levels, genus and family. At each level, all genomes of a group stayed in the same fold, so no test genome had a relative from its genus or family in training (Roberts et al., 2017). Many microbial traits are shared by close relatives, so a model can look accurate because it recognizes relatives of the training genomes (Goberna & Verdú, 2016). Grouping by genus or family removes this shortcut. The pretrained sequence models, however, may still have seen test genomes. We compared each model with three baselines. Curated marker rules call a trait from its known genes. A frequency baseline gives every genome the trait's overall frequency. A taxonomic baseline gives each test genome the share of positive training genomes in the lowest taxonomic rank that has training genomes, for example the family in genus-level cross-validation, or the overall share where no relative

exists. A random split, used only for comparison, assigns genomes to folds without regard to taxonomy. We never used test folds, external genomes or newly described species for feature selection, hyperparameter tuning or the decision thresholds of a model. The external genomes and the newly described species did, however, take part in choosing between the compared models, and the external genomes, through the release rule, in deciding which categories get yes/no calls (Supplementary Note 1). We also compared RedoxKO with a nearest-neighbor rule that gives each test genome the similarity-weighted share of positives among the five training genomes with the most similar KO content (Jaccard similarity on the 852 KOs). This rule measures how much of the accuracy comes from recognizing genomes with similar gene content alone.

The primary metric was the area under the precision-recall curve (AUPRC), which remains informative when positives are rare (Saito & Rehmsmeier, 2015). We computed it for each category on the out-of-fold predictions of all five folds together and averaged it over the 14 categories with equal weight, which we call macro AUPRC. We summarized yes/no calls by the MCC. Because the MCC depends on how each model's threshold was estimated, we ranked models by AUPRC. On the newly described species, where most tests are positive, we summarized ranking by the area under the receiver operating characteristic curve (AUROC). We obtained 95% confidence intervals by resampling whole genera or families. We compared models by paired cluster-bootstrap intervals of their difference and call two models tied when this interval includes zero. Per-category tests against the taxonomic baseline use BH correction within each group of tests.

External evaluation used 562 genomes from genera absent from training, labeled from the larger dataset (Madin et al., 2020). This set tests transfer to new genera and to a second annotation source, although the two sources may draw on overlapping publications. Its macro AUPRC covers the nine categories that map onto the larger dataset, and we sampled the set to contain positives for each. The larger dataset also supplies the pathway labels, so the set is an external test for the donor and acceptor models only.

Following recommendations for validating supervised learning in biology (Walsh et al., 2021), we recorded the models to compare, the splits, the metrics and the release rule before scoring the first model. Later experiments were registered in the same way before they were run (Supplementary Table S4). The exceptions were the rule for choosing the 100 new species of the main test, the extra-trees pathway model and the analyses added last, such as the nearest-neighbor rule. We recorded every change to the registered protocol with its date and report the main changes in Supplementary Note 1. The most important change, made after an internal audit, allowed a variant to replace the final model only if its macro AUPRC was higher in cross-validation and not lower on the external genomes. The random forest replaced the first final model, RedoxKO-LX, after all other analyses had been done. Its gain in cross-validation was not significant, so the switch rested on the external genomes and on the calls on new species (Supplementary Note 1). Figure 1 summarizes the data, models and evaluations.

Feature attribution

Within each outer genus-level fold, we fitted RedoxKO on the KO features and computed TreeSHAP values (Lundberg et al., 2020) on the test genomes of that fold. We called a KO stable if it ranked among the ten most important in at least three of the five folds. We tested whether curated marker KOs were enriched among stable features with a hypergeometric test against all KOs in the feature set, with BH correction across traits. To check that an association was not driven by phylum, we re-tested it with a Cochran-Mantel-Haenszel test stratified by phylum and by genome size, in five classes of KO count, because larger genomes carry more KOs. Fisher's exact test then compared how often curated markers and other genes passed this check.

Finally, we compared decision trees of depth three or less and single-marker rules with RedoxKO by MCC on the same folds, with genus-cluster bootstrap intervals.

Newly described species

To test the models against growth experiments, we searched PubMed for species descriptions published from 2015 to 2026 that mention electron donors or acceptors. We excluded every species whose genus, species or genome occurred in the training, external or pathway data, and kept the 167 species with an annotated RefSeq genome and at least one reported growth experiment. For 143 of them, six independent LLM agents extracted each growth experiment, which we call a test, from the description paper with its source and a quoted string, without access to gene content or model output. A sheet for a human check of 50 extracted tests is released, but this check has not yet been done. The other 24 species came from an earlier hand-curated round without quoted strings, and model output already existed for some of them, so we excluded them. An audit then removed 12 extracted tests that were not real growth experiments, such as reports of stimulated growth or of reduction during fermentation. The primary set has 255 tests (number per category in Fig. 3a).

We registered the scoring protocol before prediction. After the models had scored the 143 species, we selected a primary set of 100 species, one per genus. The selection used only the test labels and favored species with tests in rare categories. It did not use the model scores. We wrote this selection rule after scoring and did not register it. We withheld the remaining 43 species from all later comparisons and used them only to test choices made on the primary set. For a registered test on autotroph-like species, we added further autotroph-like species from description papers of 2000-2026 to those among the scored new species, extracted in the same way (Supplementary Table S1). Confidence intervals come from bootstrapping description papers, because species described together are not independent. By the same procedure, we defined in advance a pathway set of all eligible species, one per genus.

After choosing the final model, we froze it and ran two further blind tests, each registered before any species was searched (Supplementary Note 1). We searched PubMed again and excluded every genus, species and genome of all earlier sets. The first test used species described in 2026. The second used description papers of lithotrophs from any year, to obtain tests of electron donors. Because only eight such species came from genera absent from all project data, the second test also included species from genera absent from the training genomes, the same distance to training as the external genomes. Tests were extracted, checked against the source text and frozen before scoring, as for the primary set.

Comparison with other tools

We scored the same growth experiments with the following tools and baselines.

- FAPROTAX 1.2.12 (Louca et al., 2016), applied to the NCBI taxonomy of each species.
- The taxonomic baseline, which gives each genome the label frequency of its relatives in training.
- A rule based on curated marker KOs.
- METABOLIC-G v4 (Zhou et al., 2022).
- The distillation step of DRAM on KOfam annotations (Shaffer et al., 2020).
- The machine-learning predictor of Microbe Decoder, queried through its public interface with KO content only (Hackmann et al., 2026).
- For oxygen use only, the BacDive aerobic model on InterProScan Pfam annotations (Jones et al., 2014; Koblitz et al., 2025).

We mapped each tool's output to our categories by a rule fixed before running the tool (Supplementary Table S12 for METABOLIC). We scored categories that a tool does not cover as negative. For calls on new species we pooled the MCC over all tests, because several categories have fewer than ten tests, whereas on the external genomes we averaged it over categories. We did not run Traitar (Weimann et al., 2016), whose phenotypes include no lithotrophic electron donors, or gapseq (Zimmermann et al., 2021), whose pathway predictions would need a separate mapping onto our categories. All tools that use KOfam annotations

received the same KOfam release as our models.

To compare RedoxKO with METABOLIC on more genomes, we reimplemented METABOLIC's HMM searches with pyhmmer. The reimplementation reproduced all of METABOLIC's category calls on the new species, and we applied it to the external genomes and the MAG-isolate pairs. We also tested three practical uses, the precision among the 20 top-ranked genomes per category, the net benefit across decision thresholds in a decision curve (Vickers & Elkin, 2006), and the accuracy of calls with a confident label. Finally, we combined both tools in a per-category logistic regression on RedoxKO's score and METABOLIC's call, fitted on the training genomes only. We registered all these analyses in advance (Supplementary Table S4).

Confidence labels

A user acting on a single genome needs to know which calls to trust. We therefore converted the scores of RedoxKO into class-conditional (Mondrian) conformal prediction sets (Angelopoulos & Bates, 2023), which we report as confidence labels. We calibrated the sets on scores that the training genomes received in family-level cross-validation. Each prediction is then labeled a confident yes, a confident no or uncertain, and in rare cases receives no label. The sets are built to contain the correct answer for at least 90% of genomes that resemble the calibration genomes. Test genomes come from new genera, so coverage on them can fall below this target. The released workflow adds these labels to the predictions of RedoxKO in the six categories that get yes/no calls.

Metagenome-assembled genomes and habitats

From GTDB release 232 (Parks et al., 2022) we paired MAGs with isolate genomes of the same species from the external and new-species sets. All MAGs had a completeness of at least 50% and a contamination of at most 10%, as estimated with CheckM2 (Chklovski et al., 2023). We chose one MAG per species at random in each completeness bin (50-70%, 70-90% and at least 90%). We predicted both genomes of a pair from nucleotide sequence with the same gene caller. For a case study on uncultured organisms, we selected 300 high-quality MAGs (at least 90% complete and at most 5% contaminated), at most one per genus. We assigned each MAG to one of five habitats (hydrothermal vents, acid mine drainage, anaerobic digesters, seawater and the human gut) by matching its isolation source against keywords chosen before the analysis. We tested five expectations about their predicted capabilities that we had recorded before the analysis, each comparing the habitat expected to be higher with one or more other habitats, in nine tests in total. We used a one-sided Fisher's exact test on the yes/no calls, or a one-sided Mann-Whitney test on the scores for Fe(II) and H_2 oxidation, which have no calls. We corrected all nine p values together with the BH method.

Comparison with large language models

We compared two large language models (LLMs) with the trained models on 96 labeled genomes. Before querying any model, we selected these genomes as a stratified random sample with a fixed seed that kept at least five positives per category where possible. For each genome, the prompt listed which panel KOs KofamScan had assigned, taken from the shared RDF resource. Each KO came with its identifier and its KEGG definition, for example K00370 with "nitrate reductase / nitrite oxidoreductase, alpha subunit", together with the number of matching proteins and the best ratio of score to threshold. The prompt asked for the probability that the organism uses each category as defined in the database. We gave each genome two prompts. In the anonymized prompt, a code replaced the organism name, and the named prompt also gave the name. Neither prompt gave the taxonomy of the organism. Comparing the two prompts shows whether a model reads the supplied gene evidence or relies on knowledge about the organism.

We ran Claude Opus 5 (claude-opus-5) on 17 and 20 September 2026 as isolated agents in

Claude Code that read only the input batch, and the prompt forbade any tool, web search or outside lookup. These agents used default sampling settings, which this interface does not expose, and made one pass per genome. We queried GPT-6 Astra (openai/gpt-6-astra) through OpenRouter on 17 September 2026 at temperature 0 with JSON output, without tools or web search. The named GPT run stopped after 84 genomes when the API budget ran out and was not repeated, so its paired comparisons use those 84 genomes. We then applied the same design to the 20 pathway traits with Claude Opus 5, using pathway definitions in place of donor and acceptor categories. Prompts, inputs and raw outputs are in the repository.

Software and computing

We computed protein embeddings on two NVIDIA T4 GPUs and ran all other steps on a single Apple M3 Pro workstation (11 cores, 18 GB). KO assignment with pyhmmer runs on CPU only. Supplementary Table S19 gives the run time per genome of each released model. Software versions and the KOfam release are pinned in the repository.

Results

A shared resource for genomic and physiological evidence

The hackathon produced a searchable genome-protein-KO resource, an assembly-linked phenotype collection, and interfaces connecting the two. Through TogoMCP, an assistant could retrieve an organism's annotated functions, inspect their definitions, and incorporate them into answers about electron donors and acceptors. The viewer made these genomic markers inspectable alongside reported physiology and evidence grades. The resulting workflow connects what an organism has been reported to use with what its genome encodes.

This evidence-retrieval capability and the prediction benchmark serve distinct purposes. A named-organism answer can incorporate published physiological observations and therefore cannot alone demonstrate prediction accuracy for uncharacterized genomes. The supervised benchmark addressed the latter question by withholding complete taxonomic groups during evaluation.

Gene content predicts donor and acceptor use in unseen genera and families

We first tested how well gene content predicts donor and acceptor use in organisms whose genus or family was not seen in training. In genus-level cross-validation, RedoxKO reached a macro AUPRC of 0.612, above the taxonomic baseline and the curated marker rules, and it also exceeded both in family-level cross-validation (Table 1). A model that combines many genes therefore predicted the labels better than marker rules or label transfer by taxonomy. A simpler rule that copies the labels of the training genomes with the most similar KO content, however, came close to RedoxKO in cross-validation, and the difference between the two was not significant (Table 1). Much of the accuracy in cross-validation can therefore be reached through similarity in gene content alone. On the external genomes, whose labels come from a second source, RedoxKO was clearly better than this rule. The lead of RedoxKO also did not depend on positive labels that might have been inferred from genes. When only positives backed by growth experiments were kept, RedoxKO stayed as far ahead of the taxonomic baseline as before (Supplementary Table S21). Accuracy declined as test genomes became more distant from training. The macro AUPRC of RedoxKO was highest with a random split, lower with genera held out and lowest with families held out (Supplementary Table S5). The higher score with a random split shows that close relatives in training make the model look more accurate than it is for organisms without such relatives.

Sequence representations did not clearly add to gene content (Table 1). Bacformer-large alone scored below RedoxKO, and learning a weight for each protein (attention-based MIL) did not improve on averaging the protein embeddings. RedoxKO-ens-BF, which adds Bacformer-large

to the KO learners, scored higher than RedoxKO in every fold allocation, but on the external genomes it scored lower. It was also never significantly better than RedoxKO-ens, its own KO learners (Supplementary Table S21). Whether a language model adds anything beyond gene content would need more labeled genomes to show.

The lead of RedoxKO in genus-level cross-validation depended on how the genera were split into folds. Averaged over the six fold allocations, RedoxKO and RedoxKO-LX performed about equally, and RedoxKO was significantly better in only one of them (Supplementary Tables S5 and S21).

Accuracy depends on whether a trait has defined marker genes

Macro averages hid large differences among categories (Table 2). Five categories, among them sulfate reduction, sulfur oxidation and ammonia oxidation, reached AUPRC above 0.85, whereas Fe(III) reduction, other donors and other acceptors scored low. Prevalence alone did not explain these differences. Nitrite oxidation had the fewest positives, yet its AUPRC was twenty times its prevalence (Table 2). Fe(III) reduction, by contrast, reached only 0.22 at a prevalence of 9%. The 852-KO panel lacks KOs for the MtrCAB complex for extracellular electron transfer (Shi et al., 2016), and adding them through curated HMMs or multiheme cytochrome counts did not improve the score. For common traits, a high AUPRC can go with a modest MCC, so good calls also need a threshold chosen for each category.

Table 1: Performance of donor and acceptor models. Macro AUPRC in genus- and family-level cross-validation over 14 categories, and over nine categories on the external genomes. Bold marks the best value per column. Cross-validation uses the folds of this paper (other fold allocations in Supplementary Table S5). RedoxKO was chosen after the external test, so its external value is not blind.

Model	Features	Genus-level CV	Family-level CV	External genomes
Baselines				
Frequency baseline	none	0.185	0.178	0.183
Curated marker rules	marker KOs	0.317	0.317	0.338
Taxonomic baseline	taxonomy	0.472	0.380	0.380
Nearest-neighbor rule	852 KOs	0.572	0.531	0.436
Sequence models				
ESM-2	ESM-2	0.544	0.497	0.421
Attention-based MIL	ESM-2	0.548	0.495	0.405
Bacformer-large	Bacformer	0.577	0.531	0.440
KO models				
RedoxKO	852 KOs	0.612	0.562	**0.555**
RedoxKO-LX	852 KOs	0.576	0.560	0.511
RedoxKO-ens	852 KOs	0.602	0.573	0.504
Combined models				
RedoxKO-ens-ESM	KOs, ESM-2	0.616	0.584	0.483
RedoxKO-ens-BF	KOs, Bacformer	**0.625**	**0.585**	0.499

Table 2: Performance of RedoxKO by category. AUPRC, taxonomic baseline and MCC are from genus-level cross-validation. The release rule uses family-level cross-validation (Supplementary Table S6).

Category	Positives (prevalence)	AUPRC	Taxonomic baseline	AUPRC, family-level CV	MCC	Output
Sulfate reduction	63 (18%)	0.915	0.835	0.878	0.80	yes/no
Sulfur oxidation	131 (37%)	0.896	0.708	0.860	0.69	yes/no
H_2 oxidation	208 (58%)	0.880	0.814	0.877	0.50	score
Methanogen/acetogen	75 (21%)	0.870	0.821	0.858	0.75	yes/no
Ammonia oxidation	32 (9%)	0.869	0.653	0.700	0.74	yes/no
O_2 respiration	177 (49%)	0.798	0.770	0.764	0.76	yes/no
Nitrate reduction	102 (28%)	0.688	0.507	0.595	0.49	score
Nitrite oxidation	12 (3%)	0.672	0.050	0.553	0.38	score
Sulfur reduction	50 (14%)	0.584	0.399	0.622	0.36	yes/no
CO oxidation	45 (13%)	0.503	0.350	0.504	0.57	score
Fe(II) oxidation	43 (12%)	0.436	0.259	0.237	0.36	score
Fe(III) reduction	32 (9%)	0.223	0.184	0.135	0.10	score
Other donors	23 (6%)	0.135	0.081	0.161	0.19	score
Other acceptors	23 (6%)	0.104	0.180	0.120	0.04	score

Key features match known marker genes where KEGG has them

To test whether the models detect metabolic genes or merely recognize relatives, we computed TreeSHAP attributions on genera absent from training, for RedoxKO and for random forests trained on the pathway traits. Curated marker genes were significantly enriched among the stable features in 4 of 14 donor and acceptor categories and in 9 of 20 pathway traits (Fig. 2a and Supplementary Fig. S1). The two iron categories could not show this enrichment, because they have no marker among the KEGG features. The stable features were mostly known pathway genes. For nitrate reduction they included *narJ*, *narI*, *norB*, *napB* and, below the top five, *nosZ* (Fig. 2b). For sulfur oxidation they included *sox* genes and *sqr*, and for ammonia oxidation the ammonia monooxygenase subunits *amoB* and *amoC* (not shown in Fig. 2b). For sulfate reduction, they also included *dsrM*, *dsrK* and, below the top five, *dsrO*, which were missing from our list of marker genes. These genes encode subunits of the membrane complex DsrMKJOP, which reduces the DsrC trisulfide formed by sulfite reductase (Santos et al., 2015).

Two kinds of category behaved differently. The model predicted oxygen respiration mainly from the absence of genes typical of methanogens and other strict anaerobes (*hdrA2*, *hdrB2*, *mvhD*, *cdhE*), and less from genes of terminal oxidases. It therefore relies on a correlate of aerobic respiration. Oxygen use can also be predicted from genome features without functional annotation (Flamholz et al., 2024), and such predictions support analyses of oxygen adaptation across bacteria (Davín et al., 2025). Our attributions suggest why these approaches work even without relying on oxidase genes. Iron oxidation and reduction, like the manganese, arsenate and fumarate pathways, have no KO in our list of curated markers, consistent with their low accuracy (Table 2 and Supplementary Table S15).

Most of these associations did not hold within phyla. Only 15 of the 59 stable features stayed associated with their trait after controlling for phylum and genome size. Curated markers held up significantly more often than other genes (10 of 17 against 5 of 42). The associations that survive phylum control therefore mostly reflect known biology. The others may reflect shared ancestry, or too few genomes per phylum to test them.

Comparison with simple rules showed where learning adds accuracy (Fig. 2c). For ammonia oxidation, nitrate reduction and methanogenesis and acetogenesis, a single-marker rule (*amoB*, *narG* or *napA*, or a methanogenesis or acetogenesis marker) matched or exceeded RedoxKO on genera absent from training. Where no single gene defines the trait, the model was better, most clearly for hydrogen oxidation against a hydrogenase rule and for oxygen respiration against a terminal-oxidase rule, and also for sulfate reduction against *dsrB* alone. Decision trees of depth three or less matched RedoxKO in the categories with clear marker genes, but were significantly worse for oxygen respiration and sulfur reduction.

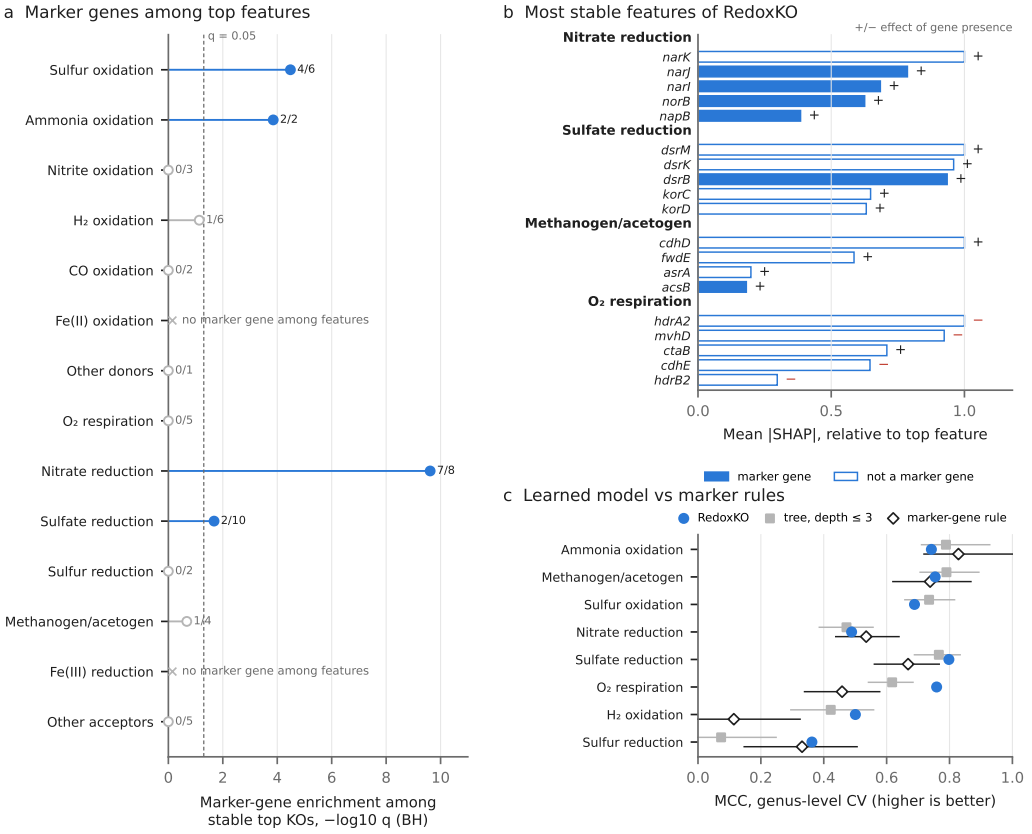


Figure 2: Top features and their agreement with known markers. (a) Marker genes among the most stable features of RedoxKO (for example 4/6). Grey points are not significant. (b) Most stable features. (c) MCC of RedoxKO, trees of depth three or less and single-marker rules, with 95% genus-bootstrap intervals of the difference from RedoxKO. Sulfur oxidation has no single-marker rule.

Predictions transfer to unseen genera of the larger dataset

We next tested whether the models transfer beyond the training data. On the external genomes, RedoxKO again clearly outperformed the taxonomic baseline (Table 1), also on a subset with balanced category sizes. The training genomes were chosen as autotrophs, but only about half of the external genomes and new species carry a carbon-fixation or methanogenesis module that is at least 80% complete (Supplementary Table S2). RedoxKO ranked the other genomes about as well, and on newly described autotroph-like species it ranked growth experiments at least as well as on all new species (Supplementary Table S2). This result depends, however, on how categories and missing values were matched between the two datasets. For the species in both datasets, the two label sources agreed moderately, mainly because they differ for hydrogen oxidation (Supplementary Note 3).

On the external genomes, RedoxKO scored highest (Table 1), also above RedoxKO-LX

(Supplementary Table S21), and the margin over RedoxKO-LX was the same for genomes whose labels do not come from FAPROTAX. RedoxKO-ens and RedoxKO-ens-BF, which had scored as high in cross-validation, did not keep this level on the external genomes. Because RedoxKO scored higher on the external genomes and gave more accurate calls on new species, it became the final model in place of RedoxKO-LX, although the two were tied in cross-validation (Supplementary Note 1). It runs on a CPU, and because each of its features is a gene, every prediction can be traced to genes (Fig. 2).

RedoxKO-ens-BF ranked the growth experiments of newly described species about as well as RedoxKO. It ranked the external genomes less well and needs protein embeddings computed on a GPU, so we release it only as an alternative for users who already have Bacformer-large vectors.

Models rank growth experiments of new species moderately well

Cross-validation removes close relatives from training but reuses the label source of the training data. We therefore tested the models against published growth experiments on newly described species, each from a genus absent from all project data. RedoxKO and RedoxKO-ens-BF ranked these tests with similar AUROC (Table 3). On all 143 scored species, RedoxKO reached an AUROC of 0.73, and 0.76 without the three categories with fewer than ten tests.

Accuracy and MCC are poor guides on this test set, because most tests are positive and the share differs between categories. A rule that gives each category its most common outcome in this test set (test-set majority) reached a higher accuracy and MCC than any model (Table 3). This rule uses the test labels, so no one could apply it to a new genome. We therefore compare models mainly by AUROC, which does not depend on a threshold or on the share of positive tests. Bacformer-large had the highest MCC among the models but a lower AUROC than the KO-based models. Its yes/no calls benefited from a threshold that happened to suit the many positive tests in this set, whereas its ranking, which does not depend on a threshold, was weaker. We therefore kept the KO-based models, which ranked tests better and whose predictions can be traced to genes.

The calls were conservative. Positive calls of RedoxKO were mostly correct (86%), whereas negative calls were often wrong (56% correct). Errors were concentrated in a few categories, above all iron reduction (Fig. 3a). RedoxKO returns only a score for this category, so these errors would not be issued as yes/no calls. Hydrogen oxidation passed the first three conditions of the release rule, but its calls were poor on the external genomes and on new species (Fig. 3), and on the external genomes RedoxKO called most negative genomes positive. It failed the fourth condition, which we added after all other results were known, so the workflow returns only a score for it. When only the six categories in which RedoxKO gives calls were counted (five of them have tests), 72% of its calls were correct. Its MCC was slightly lower than that of METABOLIC and DRAM, but neither difference was significant (Supplementary Table S21).

Table 3: Tests on new species. Performance on 255 growth experiments of 100 new species. AUROC is averaged over categories. The reference rules give one fixed answer per category. Training frequency calls a category present when at least half of the training genomes have it. Bold marks the best model per column.

Model	AUROC	Accuracy	MCC	Sensitivity	Specificity
RedoxKO-ens-BF	**0.746**	66.3%	0.370	56.9%	80.4%
RedoxKO	0.735	67.8%	0.420	55.6%	86.3%
RedoxKO-LX	0.732	64.7%	0.364	51.6%	84.3%
RedoxKO-ens	0.705	64.3%	0.374	49.0%	**87.3%**
Bacformer-large alone	0.679	**69.4%**	**0.444**	**58.2%**	86.3%
ESM-C alone	0.667	63.1%	0.290	56.9%	72.5%
Reference rules					
Training frequency	0.500	45.1%	0.128	11.1%	96.1%
Test-set majority	0.500	74.1%	0.451	83.0%	60.8%

Comparison with existing tools

We scored seven tools and baselines on the same 255 growth experiments (Table 4). The models outperformed taxonomy-based tools, which depend on close references (Douglas et al., 2020; Goberna & Verdú, 2016). FAPROTAX, for example, recognized only about a quarter of the species, and even on those its calls were less accurate than those of RedoxKO. The rule-based tools did at least as well as RedoxKO. METABOLIC gave the most accurate yes/no calls when all tests were pooled and DRAM when accuracy was averaged over categories, but none of these differences was significant. The same held in the categories in which RedoxKO gives calls, where no single category differed significantly (Fig. 3a). METABOLIC was ahead mainly in sulfur reduction, and also in hydrogen oxidation, for which RedoxKO returns only a score.

On the 43 withheld species, the only set that played no part in choosing the final model, the picture was similar. METABOLIC again had the more accurate pooled calls and RedoxKO the more accurate calls averaged over categories, and neither difference was significant. RedoxKO ranked these tests less well than those of the primary set, which may be chance in this small set (Table 4).

Two blind tests, run after the final model was frozen, gave a similar picture (Supplementary Table S20). RedoxKO ranked the tests of new lithotrophic species about as well as those of the primary set. Its ranking of electron donors was uncertain, because the description papers reported few negative donor tests. Hydrogen and sulfur oxidation were ranked well, but carbon monoxide oxidation, with only one negative test, was not. Species described in 2026 gave tests almost only for oxygen respiration, which RedoxKO ranked well. On the lithotrophic species, RedoxKO-LX ranked the tests about as well and made slightly more accurate calls, whereas on the withheld species RedoxKO ranked the tests better. The sets that played no part in choosing the final model therefore show no consistent advantage of either model.

Of the other learned predictors, Microbe Decoder made less accurate calls than RedoxKO on its three categories and ranked the tests less well, and the BacDive aerobic model ranked oxygen tests as well as RedoxKO.

RedoxKO's scores were also useful alongside METABOLIC's calls. Among the tests that METABOLIC called positive, RedoxKO's scores separated the true from the false positives. We first saw this on the 100 species and then confirmed it on the 43 withheld species, in a test registered with RedoxKO-LX and repeated with RedoxKO. Ranking tests by METABOLIC's call and then by RedoxKO's score worked better than METABOLIC alone, but not better than RedoxKO alone (Supplementary Table S4).

Table 4: Comparison with other tools. 255 growth experiments of 100 new species and, in the last group, 75 of the 43 withheld species. MCC is pooled over tests or averaged over categories, uncovered ones counted as negative. Dashes mark values not computed, because a tool gives only yes/no calls or covers too few categories. Notes give METABOLIC minus RedoxKO in MCC (Supplementary Table S21). Bold marks the best primary value per column among tools scored on all tests.

Tool	Input	MCC	MCC, category mean	AUROC	Note
Learned models					
RedoxKO	KOs	0.420	0.290	0.735	
RedoxKO-ens-BF	KOs, Bacformer	0.370	0.288	**0.746**	
Microbe Decoder	KOs	0.228	—	—	on its 3 categories MCC 0.51, AUROC 0.81 (RedoxKO 0.66, 0.94)
BacDive (oxygen)	Pfam domains	0.470	—	0.976	oxygen only, RedoxKO MCC 0.665, AUROC 0.985
Rule-based tools					
METABOLIC-G v4	curated HMMs	**0.519**	0.284	—	+0.10
DRAM	KOs	0.404	**0.359**	—	covers 8 of 12 categories
Marker-KO rule	marker KOs	0.237	0.257	0.650	
Taxonomy-based tools					
FAPROTAX 1.2.12	taxonomy	0.194	0.215	—	recognizes 27% of species, on them MCC 0.30 (RedoxKO 0.65)
Taxonomic baseline	taxonomy	0.122	0.064	0.612	
43 withheld species					
METABOLIC-G v4	curated HMMs	0.555	−0.058	—	+0.07
RedoxKO	KOs	0.488	0.153	0.619	
RedoxKO-LX	KOs	0.488	0.215	0.504	

On the external genomes, the two tools made different kinds of errors. RedoxKO gives a score and METABOLIC only yes/no calls, so only RedoxKO can rank genomes. When only the yes/no calls of the two tools were compared, RedoxKO made fewer false-positive calls and METABOLIC found more of the true positives (Fig. 3b). Averaged over the nine categories, the two tools were about equally accurate when false and missed positives were weighted equally. RedoxKO was ahead when false positives counted more, and in the categories in which it gives calls (Supplementary Table S9). These genomes took part in choosing RedoxKO and its call categories, and their labels come from the larger dataset, so the new species, tested by growth experiments, remain the stricter test.

Three checks tested the limits of these results (Supplementary Table S4). First, RedoxKO kept its advantage on the 170 external genomes whose labels do not come from FAPROTAX. These labels cover only five acceptor categories, so this check cannot rule out an effect of FAPROTAX labels on the donor categories. Second, on new species the accuracy of RedoxKO's calls tended to fall with distance from the training genomes, measured as one minus the largest Jaccard similarity of KO profiles, whereas METABOLIC's did not. The change for each tool alone was not significant, but the difference between the two tools was. Third, on incomplete MAGs the stability of the calls did not differ significantly between the two tools.

When the top-ranked external genomes in each category were chosen for testing, RedoxKO's choices contained more positives than genomes chosen from METABOLIC's positive calls. Still, about a third of them would test negative, so the scores help to shortlist genomes for testing but do not replace it. RedoxKO also gave the higher net benefit across screening thresholds on the external genomes, but not on new species. A model combining both tools did better than METABOLIC on the external genomes, but on new species it did not improve on either tool (Supplementary Table S4).

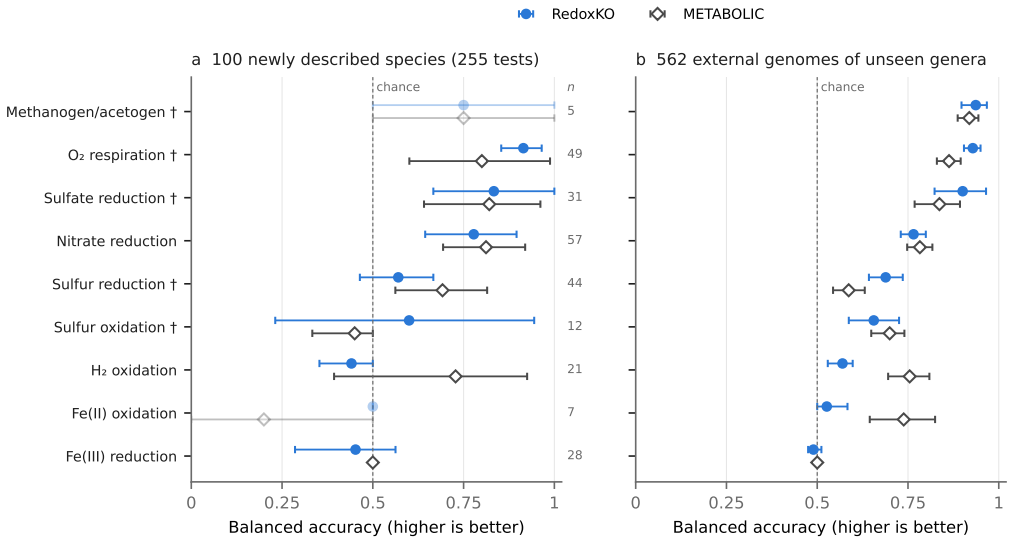


Figure 3: RedoxKO and METABOLIC by category. Balanced accuracy, the mean of sensitivity and specificity, on (a) 100 new species and (b) 562 external genomes. Bars are 95% bootstrap intervals over description papers (a) or genomes (b). † marks categories in which RedoxKO gives yes/no calls, and lighter rows have fewer than ten tests (*n*, the number of tests in a). Categories with fewer than two tests or without labels are not shown. The dashed line marks random guessing. Values for (b) are in Supplementary Table S9.

Confidence labels mark the more reliable calls

We tested whether the confidence labels keep their coverage on genomes the models have not seen, in the categories that get yes/no calls (Supplementary Table S10). On new species, the labels reached the 90% target, with lower coverage for negative than for positive tests. On the external genomes, whose labels come from the larger dataset, coverage was slightly below the target, mainly because of negative labels. Half to two thirds of the predictions (49% to 67%) received a confident label. On new species these were more often correct than the yes/no calls, but on the external genomes they were not.

For the unlabeled genomes of the focused collection, the workflow returns calibrated scores, yes/no calls where the release rule allows them, and confidence labels. Because predictions are less reliable for distant organisms, each output also states whether the genome's genus and family occur among the training genomes, as PICRUSt2 does with its distance to reference genomes (Douglas et al., 2020). These predictions help decide what to test in the laboratory, but only growth experiments can confirm them.

MAGs give nearly the same predictions as isolates

Many organisms are known only from MAGs, which often miss part of the genome. Genome-based trait predictors have been applied to such genomes (Weimann et al., 2016; Weissman et al., 2021), and we checked how much the missing genes change the predictions. We

compared the predictions for each MAG with those for an isolate genome of the same species (Supplementary Table S11). The MAG and the isolate received nearly the same calls at every completeness level, and agreement was only slightly lower below 90% completeness. There, about one in ten positive calls of the isolate was lost in the MAG. EnergyKO-M agreed most often, also for isolates outside its training set. MAGs of at least medium quality by the MIMAG standard, that is, at least 50% complete (Bowers et al., 2017), can therefore be used much like isolates, but a missing call in an incomplete MAG is weaker evidence than in an isolate.

To illustrate use on uncultured organisms, we predicted the donor and acceptor potential of 300 high-quality MAGs from five habitats, sampling at most one MAG per genus to limit taxonomic redundancy. A previous study addressed this question with marker genes in one environment (Anantharaman et al., 2016), and this analysis extends it to five. All five expectations recorded before the analysis were confirmed for RedoxKO, each significant after BH correction (Fig. 4). As expected, hydrogen-oxidation scores were higher in hydrothermal-vent MAGs than in seawater MAGs, and sulfur oxidation was predicted more often there than in human-gut MAGs. Iron-oxidation scores were higher in acid mine drainage than in each other habitat. Methanogens and acetogens were predicted more often in anaerobic digesters than in seawater, and oxygen respiration more often in seawater than in the gut or in digesters. Hydrogen oxidation has no calls after the change to the release rule, so its expectation was tested on scores, like Fe(II) oxidation. With RedoxKO-LX, with which the expectations were registered, four of the five held, and the difference for methanogens and acetogens was not significant. Because these expectations follow from textbook physiology, the analysis shows that the predictions make biological sense. It does not test whether each prediction is correct.

Some predictions made less biological sense. RedoxKO gave high hydrogen-oxidation scores to most gut and digester MAGs, consistent with fermentative [FeFe] and group 3 hydrogenases sharing KOs with uptake hydrogenases (Greening et al., 2016). A high hydrogen-oxidation score therefore shows only that a genome has hydrogenase genes.

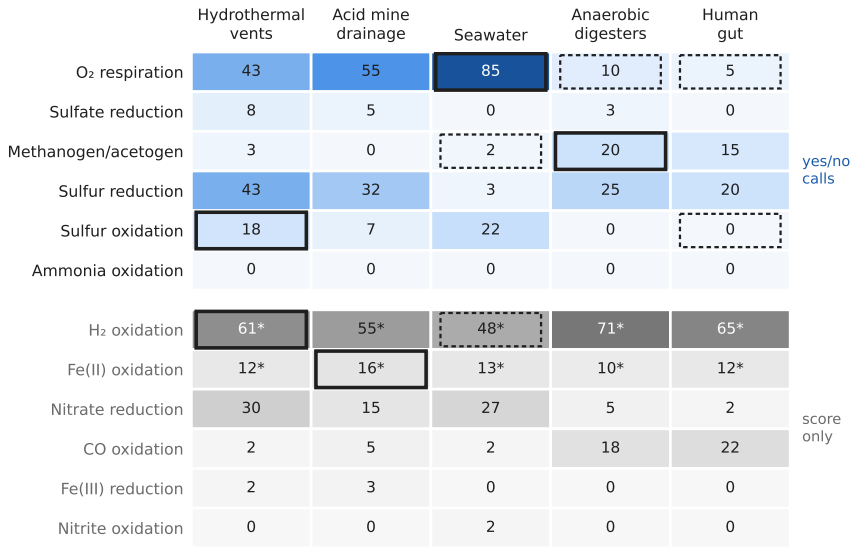


Figure 4: Predicted capabilities of 300 MAGs from five habitats. Percentage of the 60 MAGs per habitat called positive by RedoxKO (*median score ×100 where tested on scores). Blue rows have yes/no calls, grey rows scores only, cut at the cross-validation threshold. Solid outlines mark the habitat expected to be higher, dashed outlines the comparison habitat. Other donors and other acceptors are not shown.

Pathway traits are predicted moderately, and a module rule does almost as well on new species

We next tested whether the approach also predicts more specific traits, the energy-metabolism pathways. Across the 20 pathway traits, EnergyKO-M ranked genomes better than both the taxonomic baseline and a rule based on module completeness alone (Table 5). Neither another learner nor a narrower training set clearly improved it. Extra-trees, added after the protocol was registered, scored about as well in cross-validation but ranked new species less well (Table 5). Training only on autotroph-like genomes lowered accuracy slightly in cross-validation and did not change the ranking of new species significantly, also for autotroph-like species (Supplementary Table S14). Performance varied widely between traits (Supplementary Table S15). Methanogenesis and nitrate reduction were predicted best, whereas traits that lack a defined KEGG module or whose labels were largely predicted from taxonomy, such as arsenate and manganese reduction, were predicted poorly. On newly described species, EnergyKO-M and EnergyKO-BF ranked about two in three positive growth experiments above negative ones but gave weak yes/no calls (Table 5). Here the module rule ranked almost as well as the learned models, whereas the taxonomic baseline did no better than chance. No genus of these species occurs among the training genomes, and for pathway traits the higher taxonomic ranks carried little information.

Module completeness helps to explain the scores. Genomes labeled positive for a process usually carried a much more complete version of the matching KEGG module than negatives, most clearly for dissimilatory sulfate reduction and nitrification (Supplementary Table S16). The modules also separate denitrification from dissimilatory nitrate reduction to ammonium, which the donor and acceptor scheme merges into one category.

Table 5: Pathway models. Macro AUPRC over 20 traits on 3,378 genomes in cross-validation, and macro AUROC and MCC on 99 newly described species (203 growth experiments). Bold marks the best value in each column. [a]In cross-validation trained on three of five folds, because one fold was used to stop training early, so not directly comparable.

Model	Features	Genus-level CV	Family-level CV	New species AUROC	New species MCC
Learned models					
Extra-trees	KO profile	**0.506**	0.433	0.602	0.148
EnergyKO-M	KO profile + modules	0.501	0.454	0.673	0.193
EnergyKO	KO profile	0.500	**0.459**	0.650	**0.218**
EnergyKO-BF	KO profile + Bacformer	0.486	0.426	**0.701**	0.179
KO panel model	852 KOs	0.467	0.424	0.628	0.175
Bacformer, fine-tuned[a]	Bacformer-large	0.356	0.289	0.618	0.178
Baselines					
Taxonomic baseline	taxonomy	0.340	0.281	0.502	0.061
KEGG module rule	module completeness	0.213	0.213	0.660	0.154

LLMs predict well mainly when given the organism name

Finally, we tested whether LLMs can read the same gene evidence (Supplementary Table S18). Given anonymized gene evidence, both LLMs predicted donors and acceptors better than the taxonomic baseline but worse than the trained models. When the organism name was added, they reached or exceeded the trained models, and for pathways the name mattered even more. Given the organism name and no web search, an LLM therefore answers mainly from what it already knows about this organism. This is useful for summarizing existing knowledge, for example through TogoMCP, but it does not show how well an LLM would predict the

metabolism of an organism that has not yet been studied. Our experiments do not show how the LLMs used the name.

Discussion

The main outcome of the hackathon was a shared workflow connecting sequence-derived annotations with physiological evidence. The RDF resource supplies identifiable proteins, KO assignments, and functional descriptions. Literature-assisted curation supplies candidate phenotype labels and supporting statements. Visualization makes agreement and conflict inspectable, and the benchmark tests prediction beyond labeled training groups. Module annotations provide additional biological context for interpreting broad trait categories.

TogoMCP and supervised prediction have complementary roles in this workflow. TogoMCP supports flexible evidence gathering about characterized organisms, while the trained models provide reproducible scores when physiological information is unavailable. The predictor already reports, for each category, a calibrated score, a call where the category is reliable, a confidence label and whether the genus and family of the queried genome were represented in training. Exposing it as an MCP tool alongside the annotations is the next step.

Implications for model development and data curation

On newly described species, the models reached only moderate accuracy. They ranked about three in four positive growth experiments above negative ones and detected about half of the donors and acceptors found in growth experiments (Table 3). This accuracy was, however, measured in a strict test. The test genera were absent from training, and the labels came from a different source, the published growth experiments. Accuracy falls as test genomes become more distant from training (Hackmann et al., 2026), and RedoxKO itself scored higher with a random split than in genus-level cross-validation. Our estimates are therefore closer to what a user can expect for an organism that has not yet been studied.

In our tests, the model architecture mattered little. Neither the deep-learning variants we tested nor automated model search clearly exceeded RedoxKO, a random forest on the KO panel (Supplementary Table S7). This included models pretrained on large unlabeled datasets, such as fine-tuned Bacformer-large, self-supervised pretraining on unlabeled genomes and the tabular foundation model TabPFN (Hollmann et al., 2025). Instead, accuracy grew with the number of labeled genera (Fig. 5). The curve rests on four sizes and was flat between the second and third, but more labeled genomes may help more than new architectures. Label errors and missing marker genes could also limit accuracy. Genome language models trained at larger scale (Hwang et al., 2024; Nguyen et al., 2024) may also become more accurate as more labeled genomes become available.

However, additional data helped only when they matched the training data (Supplementary Table S4). The full KO profile did not rank new species better than the KO panel. Adding genomes labeled from the larger dataset made the ranking of nitrate tests on new species worse, probably because that dataset defines nitrate reduction more broadly and labels most of these genomes positive.

The labels also limit the models in a second way. A trait not reported in the literature counts as absent. We attribute to this the cautious thresholds of the models, which detected only about half of the donors and acceptors that new species used in growth experiments (Table 3). Hiding positives in the training labels barely changed the ranking, whereas hiding them in the test labels lowered the measured score (Supplementary Fig. S2), so our measured scores may underestimate how well the models rank. We tried three ways to correct this (Supplementary Table S4). First, shifting decision thresholds by the estimated reporting rate found more true positives on new species, but did not improve the calls significantly and made them worse on the external genomes. Second, positive-unlabeled bagging (Mordelet & Vert,

2014), which treats unreported traits as unknown, did not improve ranking on new species. Third, we re-read the literature for the training species with an LLM, with every claim tied to a quoted string. This resolved only a small share of the unreported traits, because species descriptions state what an organism uses but rarely what was tested and not used. None of these corrections helped. The missing negatives cannot be recovered from the existing literature, so our cross-validation scores measure how well the models predict reported traits. A possible solution is to obtain negative labels from standard growth tests, which automated cultivation platforms could run for many organisms at once (Huang et al., 2023).

On newly described species, METABOLIC's yes/no calls were at least as accurate as those of RedoxKO. Pooled over all tests METABOLIC was ahead, and averaged over categories the two were equal. Neither difference was significant, but the interval also allows a clear lead for METABOLIC. Accuracy can also make a tool that calls many tests positive look better than it is, because most published tests are positive. For example, METABOLIC called every Fe(III)-reduction test positive, which looks accurate for this reason but does not separate positives from negatives. On the external genomes, RedoxKO could rank genomes and its calls had a higher MCC, because it made fewer false-positive calls, whereas the balanced accuracy of the two tools' calls was similar.

The two kinds of tool answer different questions. A rule tells whether the genes of a pathway are present, whereas a learned score ranks genomes by how likely they are to use the pathway. Because RedoxKO's scores separated true from false positives even among METABOLIC's own positive calls, the two can be used together. Rules then provide the yes/no annotation, especially when missed positives are costly and for organisms far from the training genomes, where METABOLIC's calls lost less accuracy. Learned scores help to screen large genome collections, to decide which genomes to test first and to avoid costly false positives. RedoxKO also runs about three times faster than METABOLIC (Supplementary Table S4). Neither kind of tool, however, predicts iron and manganese metabolism well enough. For these categories, new genomic markers of extracellular electron transfer will be needed first.

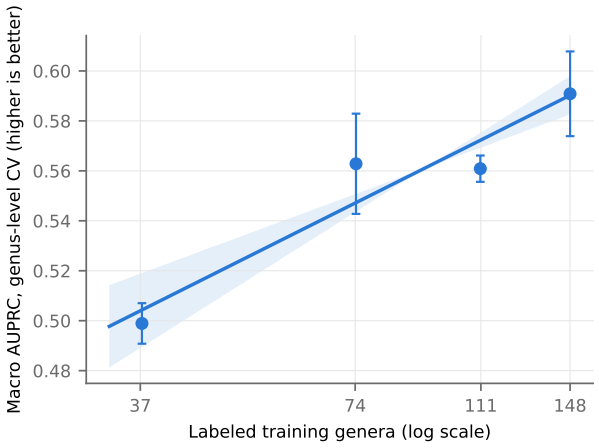


Figure 5: Relationship between accuracy and the amount of labeled data. RedoxKO in genus-level cross-validation. Points are means of three random subsamples of genera (at full size, the six fold allocations of Supplementary Table S5), with standard deviations as error bars. The blue line is a log-linear fit, with the 95% interval of its slope shaded.

Prospects for cultivation and biological discovery

Compatible donor, acceptor, and carbon fixation predictions could help prioritize substrate combinations for experimental screening. For example, evidence of sulfur oxidation together with respiratory nitrate reduction could motivate testing sulfur-containing electron donors with

nitrate as an acceptor. The relevant functions must nevertheless operate together under suitable conditions, and assimilatory transformations must be distinguished from energy-conserving respiration.

Actual medium design additionally depends on nutrient requirements, temperature, pH, substrate concentrations, toxicity, and regulation. Neither KO presence nor module completeness establishes active metabolism, and missing annotations in a MAG may reflect incomplete reconstruction rather than true gene absence. In our tests, predictions from MAGs at least 50% complete agreed closely with those from isolates, slightly less so below 90% completeness. Our focus on known autotrophs also limits extrapolation to other lifestyles. We did not experimentally validate cultivation predictions during this work. Predictions were evaluated on newly described species and on metagenome-assembled genomes, but testing defined substrate combinations in cultured strains is still necessary to establish practical utility.

Limitations

The main limitations lie in the labels. Training labels count a category not reported in the literature as negative, although most such negatives were never tested, and the literature can resolve only few of them. We attribute the cautious thresholds and low sensitivity of the models to these labels. The labeled set is also small, especially for the rarest categories (Supplementary Table S1), so results for these categories are uncertain. For most pathway traits, nearly all positives come from records that cite FAPROTAX, which assigns traits by taxon (Supplementary Table S15). Part of what the pathway models learn is therefore the taxonomy of the organism. Several external labels also come from FAPROTAX, and the larger dataset serves both as the external test for donors and acceptors and as the source of pathway labels, so these evaluations are not fully independent.

The test on new species has its own limits. The species come from papers that report which electron donors and acceptors they use, and most of their evidence comes from abstracts, so the tests may over-represent positive results. We designed the extraction so that it could not see model output, but this cannot be proven. The withheld species provide too few tests to detect anything but large differences between models (Supplementary Table S4), and the pretrained sequence models may have seen some test genomes during pretraining. RedoxKO became the final model after the tests on the external genomes and new species, so its results on these sets are not blind. Two later blind tests of the frozen model gave similar ranking, but they contained few negative tests for electron donors and showed no advantage of RedoxKO over RedoxKO-LX (Supplementary Table S20). In cross-validation, a nearest-neighbor rule on KO content came close to RedoxKO, so part of what the models learn is overall similarity in gene content. Only on the external genomes did RedoxKO clearly exceed this rule. The results of RedoxKO-LX, chosen before these tests, are given beside them (Tables 1 and 3). Finally, the predictions show which processes a genome encodes. Whether the organism carries them out in a given environment needs growth experiments to show.

A single kind of new data could solve most of these limitations. Growth experiments that record negative as well as positive outcomes, on organisms chosen from many genera, would provide both better training labels and a prospective test free of literature reporting bias.

Conclusions

During BioHackathon 2026, we developed resources linking microbial genome annotations to electron donor and acceptor information, made genomic evidence accessible through TogoMCP, and evaluated several approaches to trait prediction. Gene content alone gave most of the accuracy we observed and held up on genera and species absent from training and on metagenome-assembled genomes. Adding a genome language model gave a small gain in cross-validation that was not significant against the KO ensemble it extends and did not carry over to the external genomes. The rule-based tools METABOLIC and DRAM gave

yes/no calls at least as accurate. Pathway annotations and phenotype disagreements offered complementary routes for inspecting predictions, and the categories where prediction fails, chiefly extracellular metal reduction, show where new genomic markers and better labels are most needed. The resulting resources provide a basis for reproducible screening, continued curation, and experimental testing of genome-informed cultivation hypotheses.

Data and software availability

The manuscript repository is BH26-microbial-metabolic-traits. The microbial annotation resource was integrated with TogoMCP, whose source repository documents the service. The electron_predictor repository provides marker-matching software and an autotroph viewer.

The analysis outputs include model artifacts, metric tables, split assignments, and predictions for genomes without benchmark labels. Code, trained models, frozen cross-validation splits, evaluation sets and all metric tables are available at [GitHub URL] and archived at Zenodo ([DOI]). The released models, with SHA-256 hashes, are the donor and acceptor models RedoxKO (file RF, 36 MB), RedoxKO-LX (B2, 8.5 MB) and RedoxKO-ens-BF (B2XLE_large, 13 MB) and the pathway models EnergyKO-M (PB3, 19 MB) and EnergyKO-BF (PE). Supplementary Table S19 lists them and when to use each. A prediction script accepts RefSeq accessions or nucleotide FASTA and adds confidence labels to its output. The release also contains the genome accession lists and the test sets of newly described species, with every growth experiment, its source and quoted string. It further contains the MAG-isolate pairs and habitat MAG lists from GTDB release 232, the calls of every comparator tool, and all pre-registration documents with their hashes.

Acknowledgements and declarations

We thank the BioHackathon 2026 organizers and participants and the maintainers of the resources used here. [TO CONFIRM: author contributions, funding, and competing interests.] LLMs assisted evidence extraction, trait prediction experiments, and manuscript drafting; authors must review the final text and underlying evidence. FEDOT.MAS pipelines extracted, checked and audited the phenotype tables, with DeepSeek V4 Pro or Claude Opus 5 coordinating Gemini 3.7 Flash, Gemini 3.8 Flash or GPT-5.6 Luna. In the prediction part, LLMs were used in three ways. First, Claude Opus 5 and GPT-6 Astra were compared with the trained models, as described in the Methods (prompts, inputs and raw outputs are released). Second, LLM agents extracted the growth experiments of the newly described species, each with a quoted string from its source. Third, Claude Haiku 5.5 re-read the literature for the training species, with every claim checked against a quoted string. All reported numbers are produced by code in the released repository.

Author contributions

Shuichi Kawashima: Writing – original draft; Danil Ezhov: Methodology, Software, Formal analysis, Investigation, Validation, Data curation, Visualization, Writing – original draft (prediction sections), Writing – review & editing; Akira R Kinjo: Conceptualization, Writing – review & editing; Julia Koblitz: Conceptualization, Writing – review & editing; Takeru Nakazato: Conceptualization, Writing – review & editing; Yoko Okabbepu: Conceptualization, Writing – review & editing; Risa Otsuka: Conceptualization, Writing – review & editing.

References

Anantharaman, K., Brown, C. T., Hug, L. A., Sharon, I., Castelle, C. J., Probst, A. J., Thomas, B. C., Singh, A., Wilkins, M. J., Karaoz, U., Brodie, E. L., Williams, K. H., Hubbard, S. S., & Banfield, J. F. (2016). Thousands of microbial genomes shed light on interconnected

biogeochemical processes in an aquifer system. *Nature Communications*, *7*(1), 13219. https://doi.org/10.1038/ncomms13219 **[cito:citesAsAuthority]**.

Angelopoulos, A. N., & Bates, S. (2023). Conformal prediction: A gentle introduction. *Foundations and Trends® in Machine Learning*, *16*(4), 494–591. https://doi.org/10.1561/2200000101 **[cito:citesAsAuthority]**.

Aramaki, T., Blanc-Mathieu, R., Endo, H., Ohkubo, K., Kanehisa, M., Goto, S., & Ogata, H. (2020). KofamKOALA: KEGG Ortholog assignment based on profile HMM and adaptive score threshold. *Bioinformatics*, *36*(7), 2251–2252. https://doi.org/10.1093/bioinformatics/btz859 **[cito:citesAsAuthority]**.

Bowers, R. M., Kyrpides, N. C., Stepanauskas, R., Harmon-Smith, M., Doud, D., Reddy, T. B. K., Schulz, F., Jarett, J., Rivers, A. R., Eloe-Fadrosh, E. A., Tringe, S. G., Ivanova, N. N., Copeland, A., Clum, A., Becraft, E. D., Malmstrom, R. R., Birren, B., Podar, M., Bork, P., … Woyke, T. (2017). Minimum information about a single amplified genome (MISAG) and a metagenome-assembled genome (MIMAG) of bacteria and archaea. *Nature Biotechnology*, *35*(8), 725–731. https://doi.org/10.1038/nbt.3893 **[cito:citesAsAuthority]**.

Brbić, M., Piškorec, M., Vidulin, V., Kriško, A., Šmuc, T., & Supek, F. (2016). The landscape of microbial phenotypic traits and associated genes. *Nucleic Acids Research*, *44*(21), 10074–10090. https://doi.org/10.1093/nar/gkw964 **[cito:citesAsAuthority]**.

Breiman, L. (2001). Random forests. *Machine Learning*, *45*(1), 5–32. https://doi.org/10.1023/A:1010933404324 **[cito:citesAsAuthority]**.

Chen, T., & Guestrin, C. (2016). XGBoost: A scalable tree boosting system. *Proceedings of the 22nd ACM SIGKDD International Conference on Knowledge Discovery and Data Mining*, 785–794. https://doi.org/10.1145/2939672.2939785 **[cito:citesAsAuthority]**.

Chicco, D., & Jurman, G. (2020). The advantages of the Matthews correlation coefficient (MCC) over F1 score and accuracy in binary classification evaluation. *BMC Genomics*, *21*(1), 6. https://doi.org/10.1186/s12864-019-6413-7 **[cito:citesAsAuthority]**.

Chklovski, A., Parks, D. H., Woodcroft, B. J., & Tyson, G. W. (2023). CheckM2: A rapid, scalable and accurate tool for assessing microbial genome quality using machine learning. *Nature Methods*, *20*(8), 1203–1212. https://doi.org/10.1038/s41592-023-01940-w **[cito:citesAsAuthority]**.

Davín, A. A., Woodcroft, B. J., Soo, R. M., Morel, B., Murali, R., Schrempf, D., Clark, J. W., Álvarez-Carretero, S., Boussau, B., Moody, E. R. R., Szánthó, L. L., Richy, E., Pisani, D., Hemp, J., Fischer, W. W., Donoghue, P. C. J., Spang, A., Hugenholtz, P., Williams, T. A., & Szöllősi, G. J. (2025). A geological timescale for bacterial evolution and oxygen adaptation. *Science*, *388*(6742), eadp1853. https://doi.org/10.1126/science.adp1853 **[cito:citesAsAuthority]**.

Douglas, G. M., Maffei, V. J., Zaneveld, J. R., Yurgel, S. N., Brown, J. R., Taylor, C. M., Huttenhower, C., & Langille, M. G. I. (2020). PICRUSt2 for prediction of metagenome functions. *Nature Biotechnology*, *38*(6), 685–688. https://doi.org/10.1038/s41587-020-0548-6 **[cito:citesAsAuthority]**.

EvolutionaryScale. (2024, December 4). *ESM Cambrian: Revealing the mysteries of proteins with unsupervised learning*. https://evolutionaryscale.ai/blog/esm-cambrian **[cito:citesAsAuthority]**.

Flamholz, A. I., Goldford, J. E., Richter, P. A., Larsson, E. M., Jinich, A., Fischer, W. W., & Newman, D. K. (2024). Annotation-free prediction of microbial dioxygen utilization. *mSystems*, *9*(10), e00763–24. https://doi.org/10.1128/msystems.00763-24 **[cito:citesAsAuthority]**.

Gartlehner, G., Kugley, S., Crotty, K., Viswanathan, M., Dobrescu, A., Nussbaumer-Streit, B., Booth, G., Treadwell, J. R., Han, J. M., Wagner, J., Apaydin, E. A., Coppola, E. L., Maglione, M., Hilscher, R., Chew, R., Pilar, M., Swanton, B., & Kahwati, L. C. (2025). Artificial intelligence-assisted data extraction with a large language model: A study within reviews. *Annals of Internal Medicine*, *178*(12), 1763–1771. https://doi.org/10.7326/ANNALS-25-00739 **[cito:citesAsAuthority]**.

Geurts, P., Ernst, D., & Wehenkel, L. (2006). Extremely randomized trees. *Machine Learning*, *63*(1), 3–42. https://doi.org/10.1007/s10994-006-6226-1 **[cito:citesAsAuthority]**.

Gilemkhanov, D., Potemkin, V. A., Zaitseva, M., Revin, I., Kalyuzhnaya, A., & Nikitin, N. (2026). FEDOT.MAS: Generating multi-agent systems for complex tasks with multi-stage validation. *Proceedings of the 25th International Conference on Autonomous Agents and Multiagent Systems (AAMAS '26)*, 4077–4079. https://doi.org/10.65109/OOIG5670 **[cito:citesAsAuthority]**.

Goberna, M., & Verdú, M. (2016). Predicting microbial traits with phylogenies. *The ISME Journal*, *10*(4), 959–967. https://doi.org/10.1038/ismej.2015.171 **[cito:citesAsAuthority]**.

Greening, C., Biswas, A., Carere, C. R., Jackson, C. J., Taylor, M. C., Stott, M. B., Cook, G. M., & Morales, S. E. (2016). Genomic and metagenomic surveys of hydrogenase distribution

indicate H2 is a widely utilised energy source for microbial growth and survival. *The ISME Journal*, *10*(3), 761–777. https://doi.org/10.1038/ismej.2015.153 **[cito:citesAsAuthority]**.

Hackmann, T. J., Parris, J. P., Seshadri, R., & Lingga, C. (2026). Microbe Decoder uncovers functional traits of microbes in microbiome datasets. *Nucleic Acids Research*, *54*(W1), W160–W168. https://doi.org/10.1093/nar/gkag515 **[cito:citesAsAuthority]**.

Hollmann, N., Müller, S., Purucker, L., Krishnakumar, A., Körfer, M., Hoo, S. B., Schirrmeister, R. T., & Hutter, F. (2025). Accurate predictions on small data with a tabular foundation model. *Nature*, *637*(8045), 319–326. https://doi.org/10.1038/s41586-024-08328-6 **[cito:citesAsAuthority]**.

Huang, Y., Sheth, R. U., Zhao, S., Cohen, L. A., Dabaghi, K., Moody, T., Sun, Y., Ricaurte, D., Richardson, M., Velez-Cortes, F., Blazejewski, T., Kaufman, A., Ronda, C., & Wang, H. H. (2023). High-throughput microbial culturomics using automation and machine learning. *Nature Biotechnology*, *41*(10), 1424–1433. https://doi.org/10.1038/s41587-023-01674-2 **[cito:citesAsAuthority]**.

Hwang, Y., Cornman, A. L., Kellogg, E. H., Ovchinnikov, S., & Girguis, P. R. (2024). Genomic language model predicts protein co-regulation and function. *Nature Communications*, *15*(1), 2880. https://doi.org/10.1038/s41467-024-46947-9 **[cito:citesAsAuthority]**.

Hyatt, D., Chen, G.-L., LoCascio, P. F., Land, M. L., Larimer, F. W., & Hauser, L. J. (2010). Prodigal: Prokaryotic gene recognition and translation initiation site identification. *BMC Bioinformatics*, *11*(1), 119. https://doi.org/10.1186/1471-2105-11-119 **[cito:citesAsAuthority]**.

Ilse, M., Tomczak, J. M., & Welling, M. (2018). Attention-based deep multiple instance learning. *Proceedings of the 35th International Conference on Machine Learning*, *80*, 2127–2136 **[cito:citesAsAuthority]**.

Jones, P., Binns, D., Chang, H.-Y., Fraser, M., Li, W., McAnulla, C., McWilliam, H., Maslen, J., Mitchell, A., Nuka, G., Pesseat, S., Quinn, A. F., Sangrador-Vegas, A., Scheremetjew, M., Yong, S.-Y., Lopez, R., & Hunter, S. (2014). InterProScan 5: Genome-scale protein function classification. *Bioinformatics*, *30*(9), 1236–1240. https://doi.org/10.1093/bioinformatics/btu031 **[cito:citesAsAuthority]**.

Kanehisa, M., Furumichi, M., Sato, Y., Matsuura, Y., & Ishiguro-Watanabe, M. (2025). KEGG: Biological systems database as a model of the real world. *Nucleic Acids Research*, *53*(D1), D672–D677. https://doi.org/10.1093/nar/gkae909 **[cito:citesAsAuthority]**.

Kawashima, S., Katayama, T., Hatanaka, H., Kushida, T., & Takagi, T. (2018). NBDC RDF portal: a comprehensive repository for semantic data in life sciences. *Database (Oxford)*, bay123. https://doi.org/10.1093/database/bay123 **[cito:citesAsAuthority]**.

Kawashima, S., Okabeppu, Y., Miyazawa, S., Ichikawa, N., Nagazumi, H., Nishihara, Y., Nakazato, T., Goto, S., Kurokawa, K., Ishii, M., & Mori, H. (2026). A curated resource of chemolithoautotrophic genomes and marker genes for CO2 fixation pathway prediction. *Scientific Data*, *13*, 121. https://doi.org/10.1038/s41597-026-06655-z **[cito:citesAsAuthority]**.

Kinjo, A. R., Yamamoto, Y., Bustamante-Larriet, S., Labra-Gayo, J. E., & Fujisawa, T. (2026). TogoMCP: natural language querying of life-science knowledge graphs via schema-guided LLMs and the Model Context Protocol. *Database (Oxford)*, baag042. https://doi.org/10.1093/database/baag042 **[cito:citesAsAuthority]**.

Koblitz, J., Reimer, L. C., Pukall, R., & Overmann, J. (2025). Predicting bacterial phenotypic traits through improved machine learning using high-quality, curated datasets. *Communications Biology*, *8*(1), 897. https://doi.org/10.1038/s42003-025-08313-3 **[cito:citesAsAuthority]**.

Larralde, M. (2022). Pyrodigal: Python bindings and interface to Prodigal, an efficient method for gene prediction in prokaryotes. *Journal of Open Source Software*, *7*(72), 4296. https://doi.org/10.21105/joss.04296 **[cito:citesAsAuthority]**.

Larralde, M., & Zeller, G. (2023). PyHMMER: A Python library binding to HMMER for efficient sequence analysis. *Bioinformatics*, *39*(5), btad214. https://doi.org/10.1093/bioinformatics/btad214 **[cito:citesAsAuthority]**.

Lin, Z., Akin, H., Rao, R., Hie, B., Zhu, Z., Lu, W., Smetanin, N., Verkuil, R., Kabeli, O., Shmueli, Y., dos Santos Costa, A., Fazel-Zarandi, M., Sercu, T., Candido, S., & Rives, A. (2023). Evolutionary-scale prediction of atomic-level protein structure with a language model. *Science*, *379*(6637), 1123–1130. https://doi.org/10.1126/science.ade2574 **[cito:citesAsAuthority]**.

Louca, S., Parfrey, L. W., & Doebeli, M. (2016). Decoupling function and taxonomy in the global ocean microbiome. *Science*, *353*(6305), 1272–1277. https://doi.org/10.1126/science.aaf4507 **[cito:citesAsAuthority]**.

Lundberg, S. M., Erion, G., Chen, H., DeGrave, A., Prutkin, J. M., Nair, B., Katz, R., Himmelfarb, J., Bansal, N., & Lee, S.-I. (2020). From local explanations to global understanding with

explainable AI for trees. *Nature Machine Intelligence*, *2*(1), 56–67. https://doi.org/10.1038/s42256-019-0138-9 **[cito:citesAsAuthority]**.

Madin, J. S., Nielsen, D. A., Brbic, M., et al. (2020). A synthesis of bacterial and archaeal phenotypic trait data. *Scientific Data*, *7*, 170. https://doi.org/10.1038/s41597-020-0497-4 **[cito:citesAsAuthority]**.

Matthews, B. W. (1975). Comparison of the predicted and observed secondary structure of T4 phage lysozyme. *Biochimica et Biophysica Acta (BBA) - Protein Structure*, *405*(2), 442–451. https://doi.org/10.1016/0005-2795(75)90109-9 **[cito:citesAsAuthority]**.

Mordelet, F., & Vert, J.-P. (2014). A bagging SVM to learn from positive and unlabeled examples. *Pattern Recognition Letters*, *37*, 201–209. https://doi.org/10.1016/j.patrec.2013.06.010 **[cito:citesAsAuthority]**.

Nayfach, S., et al. (2021). A genomic catalog of Earth's microbiomes. *Nat Biotechnology*, *39*(4), 499–509. https://doi.org/10.1038/s41587-020-0718-6 **[cito:citesAsAuthority]**.

Nguyen, E., Poli, M., Durrant, M. G., Kang, B., Katrekar, D., Li, D. B., Bartie, L. J., Thomas, A. W., King, S. H., Brixi, G., Sullivan, J., Ng, M. Y., Lewis, A., Lou, A., Ermon, S., Baccus, S. A., Hernandez-Boussard, T., Ré, C., Hsu, P. D., & Hie, B. L. (2024). Sequence modeling and design from molecular to genome scale with Evo. *Science*, *386*(6723), eado9336. https://doi.org/10.1126/science.ado9336 **[cito:citesAsAuthority]**.

Parks, D. H., Chuvochina, M., Rinke, C., Mussig, A. J., Chaumeil, P.-A., & Hugenholtz, P. (2022). GTDB: An ongoing census of bacterial and archaeal diversity through a phylogenetically consistent, rank normalized and complete genome-based taxonomy. *Nucleic Acids Research*, *50*(D1), D785–D794. https://doi.org/10.1093/nar/gkab776 **[cito:citesAsAuthority]**.

Platt, J. C. (1999). Probabilistic outputs for support vector machines and comparisons to regularized likelihood methods. In A. J. Smola, P. Bartlett, B. Schölkopf, & D. Schuurmans (Eds.), *Advances in large margin classifiers* (pp. 61–74). MIT Press **[cito:citesAsAuthority]**.

Roberts, D. R., Bahn, V., Ciuti, S., Boyce, M. S., Elith, J., Guillera-Arroita, G., Hauenstein, S., Lahoz-Monfort, J. J., Schröder, B., Thuiller, W., Warton, D. I., Wintle, B. A., Hartig, F., & Dormann, C. F. (2017). Cross-validation strategies for data with temporal, spatial, hierarchical, or phylogenetic structure. *Ecography*, *40*(8), 913–929. https://doi.org/10.1111/ecog.02881 **[cito:citesAsAuthority]**.

Saito, T., & Rehmsmeier, M. (2015). The precision-recall plot is more informative than the ROC plot when evaluating binary classifiers on imbalanced datasets. *PLoS ONE*, *10*(3), e0118432. https://doi.org/10.1371/journal.pone.0118432 **[cito:citesAsAuthority]**.

Santos, A. A., Venceslau, S. S., Grein, F., Leavitt, W. D., Dahl, C., Johnston, D. T., & Pereira, I. A. C. (2015). A protein trisulfide couples dissimilatory sulfate reduction to energy conservation. *Science*, *350*(6267), 1541–1545. https://doi.org/10.1126/science.aad3558 **[cito:citesAsAuthority]**.

Sauer, D. B., & Wang, D.-N. (2019). Predicting the optimal growth temperatures of prokaryotes using only genome derived features. *Bioinformatics*, *35*(18), 3224–3231. https://doi.org/10.1093/bioinformatics/btz059 **[cito:citesAsAuthority]**.

Schober, I., Koblitz, J., Carbasse, J. S., Ebeling, C., Schmidt, M. L., Podstawka, A., Gupta, R., Ilangovan, V., Chamanara, J., Overmann, J., & Reimer, R. C. (2025). BacDive in 2025: the core database for prokaryotic strain data. *Nucleic Acids Research*, *53*(D1), D748–D756. https://doi.org/10.1093/nar/gkae959 **[cito:citesAsAuthority]**.

Shaffer, M., Borton, M. A., McGivern, B. B., Zayed, A. A., La Rosa, S. L., Solden, L. M., Liu, P., Narrowe, A. B., Rodríguez-Ramos, J., Bolduc, B., Gazitúa, M. C., Daly, R. A., Smith, G. J., Vik, D. R., Pope, P. B., Sullivan, M. B., Roux, S., & Wrighton, K. C. (2020). DRAM for distilling microbial metabolism to automate the curation of microbiome function. *Nucleic Acids Research*, *48*(16), 8883–8900. https://doi.org/10.1093/nar/gkaa621 **[cito:citesAsAuthority]**.

Shi, L., Dong, H., Reguera, G., Beyenal, H., Lu, A., Liu, J., Yu, H.-Q., & Fredrickson, J. K. (2016). Extracellular electron transfer mechanisms between microorganisms and minerals. *Nature Reviews Microbiology*, *14*(10), 651–662. https://doi.org/10.1038/nrmicro.2016.93 **[cito:citesAsAuthority]**.

Vickers, A. J., & Elkin, E. B. (2006). Decision curve analysis: A novel method for evaluating prediction models. *Medical Decision Making*, *26*(6), 565–574. https://doi.org/10.1177/0272989X06295361 **[cito:citesAsAuthority]**.

Walsh, I., Fishman, D., Garcia-Gasulla, D., Titma, T., Pollastri, G., ELIXIR Machine Learning Focus Group, Capriotti, E., Casadio, R., Capella-Gutierrez, S., Cirillo, D., Del Conte, A., Dimopoulos, A. C., Del Angel, V. D., Dopazo, J., Fariselli, P., Fernández, J. M., Huber, F., Kreshuk, A., Lenaerts, T., … Tosatto, S. C. E. (2021). DOME: Recommendations for

supervised machine learning validation in biology. *Nature Methods*, *18*(10), 1122–1127. https://doi.org/10.1038/s41592-021-01205-4 **[cito:citesAsAuthority]**.

Weimann, A., Mooren, K., Frank, J., Pope, P. B., Bremges, A., & McHardy, A. C. (2016). From genomes to phenotypes: Traitar, the microbial trait analyzer. *mSystems*, *1*(6), e00101–16. https://doi.org/10.1128/mSystems.00101-16 **[cito:citesAsAuthority]**.

Weissman, J. L., Hou, S., & Fuhrman, J. A. (2021). Estimating maximal microbial growth rates from cultures, metagenomes, and single cells via codon usage patterns. *Proceedings of the National Academy of Sciences*, *118*(12), e2016810118. https://doi.org/10.1073/pnas.2016810118 **[cito:citesAsAuthority]**.

Wiatrak, M., Abelson, D. C., Mikhaylenko, R., Viñas Torné, R., Ntemourtsidou, M., Dinan, A., Arora, D., Horsfield, S. T., Lees, J., Brbić, M., Weimann, A., & Floto, R. A. (2025). A contextualised protein language model reveals the functional syntax of bacterial evolution. *bioRxiv*. https://doi.org/10.1101/2025.07.20.665723 **[cito:citesAsAuthority]**.

Zhou, Z., Tran, P. Q., Breister, A. M., Liu, Y., Kieft, K., Cowley, E. S., Karaoz, U., & Anantharaman, K. (2022). METABOLIC: High-throughput profiling of microbial genomes for functional traits, metabolism, biogeochemistry, and community-scale functional networks. *Microbiome*, *10*(1), 33. https://doi.org/10.1186/s40168-021-01213-8 **[cito:citesAsAuthority]**.

Zimmermann, J., Kaleta, C., & Waschina, S. (2021). gapseq: Informed prediction of bacterial metabolic pathways and reconstruction of accurate metabolic models. *Genome Biology*, *22*(1), 81. https://doi.org/10.1186/s13059-021-02295-1 **[cito:citesAsAuthority]**.

Appendices

The following material supports the prediction sections. Code, result files and all pre-registration documents are in the repository.

Supplementary Note 1. Deviations from the registered protocol

We registered the donor and acceptor protocol on 17 September 2026, before scoring the first model, and followed it with the changes described below.

Additional sensitivity analyses. On 17 September, after the first internal audit and before any sequence model was scored, we added four sensitivity analyses. The audit had found likely label errors among the negatives, so one analysis treats a genome's donor or acceptor labels as missing when all of them are negative. Because the registered folds used NCBI taxonomy, a second analysis rebuilds the folds from GTDB taxonomy. The other two add five further fold allocations and keep only genomes of high assembly quality. At the same time, we decided to report two models as tied when the interval of their paired difference includes zero. We also decided to base the main claims on macro averages and to correct for multiple testing with the Benjamini-Hochberg method within each group of tests. A second macro average, over 12 categories, leaves out other donors and other acceptors. The 14-category average remains the main measure. These additions did not change any registered analysis. They were run with RedoxKO-LX, the final model at the time, which stayed ahead of the taxonomic baseline in all of them. Supplementary Table S5 gives the results.

Choice of the final model. After the second internal audit, on 18 September, we added a rule for replacing the final model, then RedoxKO-LX. A new model replaces it only if it scores higher in cross-validation and does not score lower on the external genomes in any reported subset. We wrote this rule together with the external test. RedoxKO-ens scored higher in cross-validation but failed the rule, because it scored significantly lower on the external genomes whose labels do not come from FAPROTAX (Supplementary Table S21). On all external genomes the two were tied. We then applied the same check to all registered models. Bacformer-large alone scored about as well as RedoxKO-LX in genus-level cross-validation, lower at family level and significantly lower on the external genomes. We added this step after the cross-validation ranking was known, although the external comparison had already been planned and computed. It changed which model became the final model, but not any reported value.

Switch to the random forest. We had trained a random forest on the same 852 KOs as a comparison model but had not applied this rule to it. On 21 October, after all other analyses had been done, we applied the rule to it. In genus-level cross-validation the random forest scored higher than RedoxKO-LX, but the interval of the difference included zero, so by our definition the two were tied (Supplementary Table S21). The switch therefore rested on two results. The random forest scored significantly higher on the external genomes, also on those whose labels do not come from FAPROTAX, and its yes/no calls on the 100 primary new species were significantly more accurate. The 43 withheld species were not used. It became the final model, RedoxKO, and the former final model was renamed RedoxKO-LX. We then repeated every analysis of the final model with the random forest. Because the switch was made after all test results were known, the results of RedoxKO on the external genomes and new species are not blind. The experiments registered in advance had been run with RedoxKO-LX. We repeated them with RedoxKO and report the repeats in Supplementary Table S4. The original results remain in the repository.

Fourth condition of the release rule. On 22 October we added a fourth condition to the release rule: where the external genomes have labels for a category, the MCC of its calls on them must exceed 0.2. Only hydrogen oxidation failed it (Supplementary Table S6), so it now returns a score only, and its habitat expectation was tested on scores, as the registration prescribes for categories without calls. We made this change after all results were known.

Blind tests of the frozen model. On 22 October we froze RedoxKO with its release rule, recorded the hashes of the model files and registered two blind tests before searching for species (Supplementary Table S20). The first used species described in 2026. Its registered search window, 15 September to 22 October, gave no eligible species, so before extraction we widened it to the whole year. The second used description papers of lithotrophs from any year. Only eight species came from genera absent from all project data, so before extraction we added species from genera absent from the training genomes. In both tests we corrected the quote check and the rule for statements that are not growth experiments before scoring. After scoring we restricted the yes/no calls to the six categories that get them, as registered. Both tests are reported regardless of their outcome.

Pathway task. On 18 September we added the pathway task with its own registration, labels, genomes and frozen folds. Nothing in the donor and acceptor analysis was rerun or chosen again because of it.

Alternative models. On 21 and 22 September, after the test on newly described species, in which RedoxKO-ens-BF and EnergyKO-BF did not differ significantly from the final models, we also named these two as alternative models. The final models did not change.

LLM comparison. We did not keep the transcripts of the isolated Claude agents that ran on 17 and 20 September, so only their final answers are available.

Further experiments. From 14 to 21 October we registered further experiments before running them. Their results are reported in Supplementary Table S4 regardless of their outcome.

Supplementary Note 2. Audit of the phenotype collection

In October 2026 we checked the phenotype collection against its cited sources and classified the errors as in a study of LLM-assisted data extraction (Gartlehner et al., 2025). Supplementary Table S3 lists them. In the separate donor and acceptor table, nearly all errors concerned the format of quotes. After the corrections, we re-checked a stratified random sample of 300 items against the sources. Of the 239 sampled values graded A to C that could be checked against an open source, 2.1% still had a major error and 18.0% had an error of any kind. Gartlehner et al. found major errors in 2.5% of LLM-assisted and 2.7% of human-only extractions. Every error found in the sample was corrected. The audit changed 17 of the 3,828 training labels that it could match by name. Re-scoring the cross-validation predictions with the corrected labels changed macro AUPRC by at most 0.004.

Supplementary Note 3. Agreement between the two label sources

Comparing the two label sources shows how much the evaluations depend on the labels. For the species in both, the labels of the phenotype collection and the larger dataset agreed moderately, and RedoxKO agreed with the larger dataset slightly less (Supplementary Table S17). Most of the disagreement came from hydrogen oxidation, which the phenotype collection assigned to far more species than the larger dataset. Without it, the two sources agreed well. The sources therefore differ in what they label and in how much evidence they cover. They are also not fully independent, because most of the shared species have records from FAPROTAX in the larger dataset, which assigns traits from the organism's taxonomy.

The comparison also found genomes labeled negative for a trait although their marker genes or other records indicated its presence. For example, five genomes with a complete methyl-coenzyme M reductase (*mcrABG*) were labeled as non-methanogens, and nine strictly aerobic ammonia or nitrite oxidizers were labeled as not respiring oxygen. Such cases are candidates for re-curation with the viewer and other sources. A prediction supported by marker genes does not prove a labeling error, because gene regulation, pathway direction, growth conditions and strain differences can also explain a disagreement.

Supplementary tables

Square brackets give 95% bootstrap confidence intervals, resampled over genera or families in cross-validation, over genera for the external genomes and over the description papers for new species.

Table S1: Datasets used in this study. All genomes are isolates except the MAGs.

Dataset	Source and selection	Genomes	Labels	Use
Training				
Donor/acceptor	Phenotype collection (this study), extracted from published papers, focused set of autotrophs	358	14 categories, 12-208 positives per category (3-58%)	Cross-validation
Pathways	Larger dataset (Madin et al., 2020), species with pathway labels, one genome per species	3,378	20 traits, 30-1,814 positives per trait (1-54%)	Cross-validation
Evaluation				
External genomes	Larger dataset, genera absent from donor/acceptor training	562	9 categories, 19-289 positives per category (3-62% of the genomes labeled for it)	Test on unseen genera
New species	Described 2015-2026, genus absent from all project data, one per genus in the primary set of 100, and 43 withheld species	143	330 experiments, 61% positive	Test on new species
New species, pathways	Same selection	99	203 experiments, 53% positive	Test on new species
New species, autotroph-like	70 autotroph-like species of the 143 plus 29 taken from description papers of 2000-2026, extracted the same way	99	220 experiments	Registered autotroph test
MAG-isolate pairs	GTDB release 232 (Parks et al., 2022), MAGs $\geqslant$ 50% complete and $\leqslant$ 10% contaminated, one per species and completeness bin, paired with 222 isolates	372 MAGs	Isolate predictions	Effect of completeness

Dataset	Source and selection	Genomes	Labels	Use
Applications and resources				
Habitat MAGs	GTDB release 232, ⩾ 90% complete and ⩽ 5% contaminated, one per genus, five habitats	300	None, uncultured	Habitat comparison
Unlabeled genomes	Focused collection without labels	228	None, no published evidence	Predictions
LLM subset	Stratified sample of training genomes	96	14 categories, 20 traits	Comparison with LLMs
Shared KO resource	KofamScan annotations of RefSeq genomes, available through TogoMCP	23,434	KO annotations only	Annotation source

Table S2: Performance of RedoxKO on genomes with and without a carbon-fixation pathway. Autotroph-like genomes have a KEGG module for carbon fixation or methanogenesis at least 80% complete. [a]Collected for this test, 220 growth experiments, MCC 0.41.

Test set	Autotroph-like genomes	Macro AUROC, autotroph-like	Macro AUROC, other	Difference
External genomes	285 of 562 (51%)	0.82	0.86	−0.04
New species, primary set	46 of 100 (46%)	0.83	0.78	+0.05
New species, autotroph-like set[a]	99	0.76	—	—

Table S3: Errors found in the audit of the phenotype collection. In round brackets, the share of the 6,880 values in the collection of October 2026, counting each compound separately. The 1,817 records of the September snapshot cover only donors and acceptors. Severity classes follow the scheme of (Gartlehner et al., 2025). Grade D, value kept without citable support.

Error type	Example	Collection	Outcome in the collection
Major	*Record says something wrong*	***158 (2.3%)***	
Wrong organism or strain	Quote about another species described in the same paper	86 (1.2%)	78 fixed, 5 removed, 3 grade D
Wrong role	A compound the organism detoxifies taken as its energy source	26 (0.4%)	15 fixed, 3 removed, 8 grade D
Wrong kind of value	Growth limit taken as optimum, pyrite taken as Fe(II)	24 (0.3%)	23 fixed, 1 removed
Negation misread	"Not used" read as used	11 (0.2%)	9 fixed, 2 removed
Wrong value	Transcription or unit error	11 (0.2%)	8 fixed, 3 grade D
Minor	*Value kept, support wrong*	***1,955 (28.4%)***	

Error type	Example	Collection	Outcome in the collection
No citable quote	Value imported from a survey sheet or curator note without a literature quote	1,186 (17.2%)	695 fixed, 12 removed, 479 grade D
Quote does not state the value	Compound mentioned only as a medium component	316 (4.6%)	264 fixed, 15 removed, 37 grade D
Hedged evidence graded too high	"Genes for", "may" or "related strain" graded as an experiment	145 (2.1%)	139 fixed, 6 grade D
Source not a publication	Link to a chat-assistant page	124 (1.8%)	71 fixed, 4 removed, 49 grade D
Quote altered	Sentence paraphrased, shortened or merged	81 (1.2%)	81 fixed
Quote not in the cited source	Correct sentence attached to the wrong paper	55 (0.8%)	54 fixed, 1 grade D
Grade too high	Grade A resting on a single paper	40 (0.6%)	38 fixed, 2 grade D
Grade too low	A second independent source not counted	8 (0.1%)	8 fixed
Format only	*Content unchanged*	***13 (0.2%)***	
Corrupted text	Characters lost in PDF text extraction	6 (0.1%)	6 fixed
Duplicate row	Same culture under a former and a current genus name	7 (0.1%)	7 removed
Not assessed		***430 (6.2%)***	
Type not recorded	Correction by the first FEDOT.MAS validation run	375 (5.5%)	366 fixed, 9 removed
Left to the authors	Conflicting sources or a deciding source behind a paywall	48 (0.7%)	Author decision
Other	Reason given in the error log	7 (0.1%)	7 fixed
Total		**2,556 (37.2%)**	**1,920 fixed or removed, 588 grade D, 48 author decision**

Table S4: Experiments registered in advance. Differences between models or tools in the main measure, for RedoxKO unless another model is named. The experiments were registered with RedoxKO-LX and repeated with RedoxKO. Expected in advance gives the predicted outcome and, in round brackets, our probability for it. As expected is Yes when the registered success criterion was met.

Experiment	Main measure	Expected in advance	Result	95% CI	As expected
Improving RedoxKO					
Adding 546 genomes from the larger dataset (84% nitrate-positive)	Nitrate AUROC, new species	Not stated	−0.20	[−0.34, −0.07]	—
Thresholds shifted by the reporting rate	MCC, withheld species	No loss	+0.08	[−0.09, +0.23]	Yes
LLM re-reading of the training literature	Unreported cells resolved	Enough to retrain	2.4%		No
Positive-unlabeled bagging	Macro AUROC, new species	Gain unlikely (10-15%)	−0.03	[−0.11, +0.01]	Yes
No class weighting	Macro AUPRC difference, genus-level CV	None better (about 75% for all three)	−0.007	[−0.035, +0.026]	Yes

Experiment	Main measure	Expected in advance	Result	95% CI	As expected
Undersampling of negatives to 1:1	Macro AUPRC difference, genus-level CV (family level +0.007)	Lower (about 65%)	−0.025	[−0.050, +0.015]	No
Oversampling of positives to 1:1	Macro AUPRC difference, genus-level CV	None better	−0.015	[−0.037, +0.025]	Yes
Full profile of 10,517 KOs	Macro AUROC, new species	Gain unlikely (15-20%)	−0.03	[−0.10, +0.03]	Yes
METABOLIC's call, then RedoxKO's score	Macro AUROC vs RedoxKO, new species	Gain likely (55-65%)	+0.001	[−0.07, +0.07]	No
Model combining RedoxKO and METABOLIC	MCC vs METABOLIC, new species	Gain possible (25-35%)	−0.11	[−0.24, +0.02]	No
Same model, external genomes	Macro MCC vs METABOLIC	—	+0.10	[0.06, 0.14]	—
RedoxKO compared with METABOLIC					
Ranking within METABOLIC's positive calls	AUROC, withheld species	Inconclusive (power 26%)	0.82	[0.64, 1.00]	No
Ranking of external genomes	Macro AUROC difference	RedoxKO better (60-70%)	+0.09	[0.06, 0.12]	Yes
Top 20 external genomes per category	Precision difference	RedoxKO better (about 80%)	+0.27	[0.19, 0.32]	Yes
Net benefit, external genomes	Difference per threshold	Depends on the threshold	RedoxKO higher at 0.05-0.90		Partly
Calls with a confident label	Accuracy difference, new species	Even (about 50%)	+0.01	[−0.06, +0.11]	Yes
MAGs 50-70% complete	Agreement with the isolate, RedoxKO minus METABOLIC	RedoxKO more stable (50-60%)	+1.9 points	[−0.2, +4.1]	No
Speed	Seconds per genome	RedoxKO faster	14.6 vs 38-60 (3.3 for the KO scan)		Yes
Autotroph-like new species (99)	Macro AUROC, RedoxKO	At least 0.73 (about 70%)	0.764	[0.64, 0.86]	Yes
Same species	MCC, RedoxKO	Below 0.36 (about 60%)	0.406	[0.27, 0.54]	No
Validity checks					
External genomes without FAPROTAX labels	Macro AUROC difference	Uncertain (about 50%)	+0.16	[0.13, 0.18]	—
Distance from the training genomes	Accuracy loss, RedoxKO minus METABOLIC	RedoxKO loses more (55-65%)	−0.24	[−0.43, −0.04]	Yes
Pathway model on the phenotype collection only	Macro AUPRC, five traits	Similar or lower (about 60%)	−0.09	[−0.14, −0.03]	Yes

Other learners on the Bacformer-large vector	Macro AUPRC vs logistic regression	None better (about 70%)	−0.045 to −0.009	Yes

Table S5: Effect of the fold allocation on donor and acceptor results. Macro AUPRC with the folds used in the paper, five further random allocations, folds rebuilt from GTDB taxonomy, a random split without taxonomic grouping and the other sensitivity analyses of Supplementary Note 1. The last column gives the paired difference between RedoxKO-ens-BF and RedoxKO.

Folds	RedoxKO	RedoxKO-LX	RedoxKO-ens	RedoxKO-ens-BF	Taxonomic baseline	RedoxKO-ens-BF minus RedoxKO
Genus-level CV						
Folds used in the paper	0.612	0.576	0.602	0.625	0.472	+0.013
Random split, without taxonomic grouping	0.659	0.647	—	—	—	—
Five further allocations, mean	0.587	0.580	0.608	0.625	0.470	+0.038 (3 of 5 significant)
All six allocations, mean	0.591	0.580	0.607	0.625	0.470	—
GTDB-based folds	0.608	0.591	0.607	0.615	0.485	+0.007
All-negative label blocks as missing	—	0.604	—	—	0.494	—
High assembly quality only	—	0.567	—	—	0.476	—
12 categories, without other donors and acceptors	0.695	0.652	—	—	0.529	—
Family-level CV						
Folds used in the paper	0.562	0.560	0.573	0.585	0.380	+0.023
Five further allocations, mean	0.550	0.556	0.570	0.592	0.382	+0.042 (0 of 5 significant)
All six allocations, mean	0.552	0.557	0.571	0.591	0.382	—
GTDB-based folds	0.579	0.577	0.579	0.596	0.414	+0.017
All-negative label blocks as missing	—	0.579	—	—	0.403	—
High assembly quality only	—	0.556	—	—	0.387	—
12 categories, without other donors and acceptors	0.632	0.628	—	—	0.429	—

Table S6: Inputs of the release rule for RedoxKO. The four conditions are given in the Methods. A dash marks a category without external labels. Ammonia oxidation has no external labels, so its calls were checked only in cross-validation.

Category	MCC, family-level CV	95% CI	AUPRC, family-level CV	Taxonomic baseline, family-level CV	q vs taxonomic baseline	External AUPRC relative to share of positives	External MCC	Output
Sulfate reduction	0.82	[0.69, 0.91]	0.878	0.647	0.008	13.8	0.80	yes/no
Methanogen/acetogen	0.77	[0.60, 0.88]	0.858	0.695	0.006	8.6	0.75	yes/no
Sulfur oxidation	0.62	[0.48, 0.73]	0.860	0.584	0.006	5.2	0.34	yes/no
O_2 respiration	0.61	[0.41, 0.78]	0.764	0.672	0.043	1.6	0.85	yes/no
Nitrate reduction	0.52	[0.40, 0.63]	0.595	0.496	0.096	1.9	0.52	score
Ammonia oxidation	0.61	[0.30, 0.89]	0.700	0.364	0.019	—	—	yes/no
Sulfur reduction	0.41	[0.25, 0.54]	0.622	0.305	0.006	3.8	0.48	yes/no
H_2 oxidation	0.38	[0.23, 0.53]	0.877	0.714	0.006	2.5	0.09	score
CO oxidation	0.41	[0.18, 0.60]	0.504	0.332	0.028	—	—	score
Nitrite oxidation	0.44	[0.13, 0.66]	0.553	0.055	0.006	—	—	score
Fe(III) reduction	0.18	[0.05, 0.30]	0.135	0.159	0.625	0.9	−0.03	score
Other donors	0.18	[0.03, 0.30]	0.161	0.114	0.247	—	—	score
Other acceptors	0.03	[−0.09, 0.16]	0.120	0.059	0.011	—	—	score
Fe(II) oxidation	0.04	[−0.13, 0.23]	0.237	0.121	0.008	6.5	0.23	score

Table S7: Deep-learning variants and automated model search for donors and acceptors. Macro AUPRC. Bold marks the best value in each column.

Model	Genus-level CV	Family-level CV
Deep-learning variants and model search		
Multilayer perceptron on the KO panel	0.605	0.548
Multilayer perceptron, wide 24-configuration grid	0.607	0.551
Multi-task network with pathway labels	0.586	0.548
Same network without pathway labels (control)	0.605	0.561
Self-supervised pretraining on 3,400 unlabeled genomes, then fine-tuning	0.604	0.552
TabPFN v2 tabular foundation model	0.594	0.562
Automated model search (FLAML)	0.569	0.541
ESM-2 protein embeddings, average or maximum	0.544	0.497
ESM-C 300M protein embeddings, average or maximum	0.574	0.509
Attention-based multiple-instance learning over all ESM-2 proteins	0.548	0.495
Attention-based multiple-instance learning over ESM-2 proteins with energy-panel KOs	0.544	0.486
Bacformer-base genome vector	0.551	0.522
Bacformer-large genome vector	0.577	0.531
Bacformer-base, MAG weights	0.540	0.497
Bacformer-large, MAG weights	0.560	0.532

Model	Genus-level CV	Family-level CV
Bacformer-large vector with a multilayer perceptron	0.569	0.519
Bacformer-large frozen, trained head	0.556	0.532
Bacformer-large fine-tuned (top two layers)	0.549	0.535
Other learners on the Bacformer-large vector		
Random forest	0.562	0.482
Extra-trees	0.546	0.482
XGBoost	0.562	0.504
TabPFN v2	0.568	0.504
Automated model search (FLAML)	0.532	0.482
Reference models		
RedoxKO-ens	0.602	**0.573**
RedoxKO	**0.612**	0.562
RedoxKO-LX	0.576	0.560

Table S8: Class balancing by the share of positive genomes. Macro AUPRC of RedoxKO with class weights, and the change with each other treatment. In round brackets, the number of categories that scored lower. Rows marked 95% CI give the interval of the change above.

Categories (number)	Class weights	No weighting	Undersampling	Oversampling
RedoxKO, genus-level CV				
$<$ 10% positive (5)	0.401	−0.011 (3 of 5 lower)	−0.077 (3 of 5 lower)	−0.051 (2 of 5 lower)
10-30% positive (6)	0.666	−0.010 (4 of 6 lower)	+0.002 (3 of 6 lower)	+0.007 (3 of 6 lower)
$>$ 30% positive (3)	0.858	+0.004 (0 of 3 lower)	+0.007 (0 of 3 lower)	+0.003 (1 of 3 lower)
All categories (14)	0.612	−0.007	−0.025	−0.015
95% CI		[−0.035, +0.026]	[−0.050, +0.015]	[−0.037, +0.025]
RedoxKO, family-level CV				
$<$ 10% positive (5)	0.334	+0.073 (1 of 5 lower)	+0.045 (2 of 5 lower)	+0.026 (1 of 5 lower)
10-30% positive (6)	0.616	−0.017 (3 of 6 lower)	−0.018 (3 of 6 lower)	+0.001 (3 of 6 lower)
$>$ 30% positive (3)	0.834	−0.001 (2 of 3 lower)	−0.005 (1 of 3 lower)	+0.002 (1 of 3 lower)
All categories (14)	0.562	+0.019	+0.007	+0.010
95% CI		[−0.015, +0.042]	[−0.024, +0.030]	[−0.014, +0.036]
RedoxKO-LX, registered				
All categories (14), genus-level CV	0.576	+0.009	+0.019	+0.007
95% CI		[−0.008, +0.017]	[−0.015, +0.034]	[−0.006, +0.027]

Categories (number)	Class weights	No weighting	Undersampling	Oversampling
All categories (14), family-level CV	0.560	−0.007	−0.032	−0.014
95% CI		[−0.023, +0.005]	[−0.065, −0.001]	[−0.031, +0.005]

Table S9: RedoxKO and METABOLIC by category on 562 external genomes. METABOLIC's AUROC is computed from its yes/no calls and therefore equals its balanced accuracy. Bold marks a significant difference, and † marks categories in which RedoxKO gives yes/no calls. In the macro rows, sensitivity and specificity are pooled over all tests.

Category (positives)	Tool	AUROC	MCC	Sensitivity	Specificity
O_2 respiration † (289)	RedoxKO	**0.98**	**0.85**	0.91	**0.94**
	METABOLIC	0.86	0.74	0.92	0.80
Nitrate reduction (227)	RedoxKO	**0.85**	0.52	0.78	0.76
	METABOLIC	0.78	0.56	0.74	**0.82**
Sulfate reduction † (32)	RedoxKO	**0.94**	**0.80**	0.81	**0.99**
	METABOLIC	0.84	0.37	0.88	0.80
Sulfur reduction † (102)	RedoxKO	**0.91**	0.48	0.41	**0.97**
	METABOLIC	0.59	0.22	0.25	0.93
Methanogen/acetogen † (56)	RedoxKO	**0.99**	**0.75**	0.93	**0.94**
	METABOLIC	0.92	0.62	0.96	0.87
Fe(III) reduction (59)	RedoxKO	0.51	−0.03	0.02	**0.96**
	METABOLIC	0.50	0.00	**1.00**	0.00
H_2 oxidation (34)	RedoxKO	0.78	0.09	0.97	0.17
	METABOLIC	0.75	**0.25**	0.88	**0.63**
Sulfur oxidation † (48)	RedoxKO	**0.86**	0.34	0.35	**0.96**
	METABOLIC	0.70	0.22	**0.92**	0.48
Fe(II) oxidation (19)	RedoxKO	0.64	0.23	0.05	**1.00**
	METABOLIC	0.74	0.18	**0.79**	0.69
Macro, external (4,962 pairs)	RedoxKO	**0.83**	**0.45**	0.71	0.85
	METABOLIC	0.74	0.35	0.80	0.65
Macro, new species (255 experiments)	RedoxKO	0.74	0.29	0.56	0.86
	METABOLIC	0.64	0.28	0.80	0.72

Table S10: Coverage of the confidence labels of RedoxKO. Labels in the six call categories, five with external labels and new-species tests. Coverage is the share of labels that contain the correct answer (target 90%). Confident is the share labeled a confident yes or no. The last two columns give the accuracy of confident labels and of yes/no calls.

Test set	Predictions	Coverage	Positives	Negatives	Confident	Correct if confident	Calls correct
Training genomes	2,148	90.8%	91.5%	90.6%	76%	89%	89%

Test set	Predictions	Coverage	Positives	Negatives	Confident	Correct if confident	Calls correct
External genomes	2,714	87.1%	97.9%	84.5%	67%	88%	92%
New species	141	90.1%	94.6%	81.6%	49%	80%	72%

Table S11: Predictions from MAGs and from isolates of the same species. Columns give MAG completeness and, in round brackets, the number of MAGs. Calls are compared at each category's threshold for all 14 donor and acceptor categories, including score-only ones, and all 20 pathway traits. Changes in AUROC are MAG minus isolate.

Model	90-100% (189)	70-90% (132)	50-70% (51)
Calls that agree			
RedoxKO	98.8%	96.6%	96.4%
RedoxKO-ens-BF	98.0%	96.2%	95.7%
EnergyKO-M	98.9%	97.3%	96.5%
RedoxKO calls			
Cohen's kappa	0.97	0.91	0.89
Isolate positives kept in the MAG	97%	91%	87%
Change in AUROC			
RedoxKO	0.00	−0.02	−0.02
RedoxKO-ens-BF	0.00	0.00	−0.03

Table S12: Mapping of METABOLIC functions onto the categories. A category is positive when METABOLIC reports any of the listed functions.

Category	METABOLIC functions counted as a positive call
Sulfur oxidation	Sulfide oxidation or Sulfur oxidation or Thiosulfate oxidation (*dsrA* and *dsrB* not counted)
Ammonia oxidation	Ammonia oxidation
Nitrite oxidation	Nitrite oxidation
H_2 oxidation	Ni-Fe Hydrogenase
CO oxidation	Aerobic CO oxidation
Fe(II) oxidation	Iron oxidation
Nitrate reduction	Nitrate reduction
Sulfate reduction	Sulfate reduction
Sulfur reduction	Sulfite reduction or Sulfur reduction (sulfite reduction includes *dsrA* or *dsrB*)
Methanogen/acetogen	Methane production or Wood Ljungdahl pathway
Fe(III) reduction	Iron reduction or Metal (Iron/Manganese) reduction
O_2 respiration	Any aa3-, bo-, caa3- or cbb3-type terminal oxidase (bd-type excluded)

Table S13: Pathway model trained only on the phenotype collection. EnergyKO-M trained on the 174 genomes of the collection with pathway labels, compared with the final EnergyKO-M, for the five traits with enough positives.

Comparison	Collection only	Final model	Difference
Traits that can be modeled	5	20	
Macro AUPRC, genus-level CV (174 genomes, 5 traits)	0.63	0.73	−0.09
Macro AUPRC, family-level CV	0.61	0.66	−0.05
Macro AUROC, new species (137 experiments, 4 traits)	0.67	0.75	−0.08
AUPRC per trait, genus-level CV			
Nitrate reduction	0.58	0.78	−0.20
Sulfur oxidation	0.63	0.75	−0.12
Sulfur reduction	0.56	0.65	−0.09
Hydrogen oxidation	0.39	0.44	−0.05
Methanogenesis	1.00	1.00	0.00

Table S14: Pathway models trained only on autotroph-like genomes. Macro AUPRC in cross-validation on the 1,648 autotroph-like genomes and macro AUROC on new species, for the 17 traits with at least 25 positives. The last rows give the difference from EnergyKO-M, final.

Model	Genus-level CV	Family-level CV	New species, all	New species, autotroph-like
EnergyKO-M, final	0.529	0.502	0.67	0.82
EnergyKO-M, autotroph-like genomes only	0.509	0.481	0.70	0.86
Extra-trees, autotroph-like genomes only	0.509	—	0.60	0.76
Difference from EnergyKO-M, final				
EnergyKO-M, autotroph-like genomes only	−0.020	−0.020		
Extra-trees, autotroph-like genomes only	−0.020	—		

Table S15: Performance of EnergyKO-M by pathway trait. AUPRC in genus-level cross-validation. Chance level is the AUPRC of a model that only guesses, which equals the share of positive genomes. FAPROTAX share is the share of positive labels taken from FAPROTAX, which assigns traits by taxon.

Trait (positives)	EnergyKO-M	Taxonomic baseline	Chance level	FAPROTAX share
Methanogenesis (132)	0.98	0.99	0.04	24%
Nitrate reduction (1,814)	0.85	0.75	0.54	47%
Photoautotrophy (54)	0.76	0.41	0.02	100%
Sulfate reduction (100)	0.76	0.68	0.03	79%
Sulfur reduction (229)	0.73	0.58	0.07	82%

Trait (positives)	EnergyKO-M	Taxonomic baseline	Chance level	FAPROTAX share
Photoheterotrophy (77)	0.69	0.15	0.02	100%
Fermentation (465)	0.67	0.54	0.14	100%
Sulfur oxidation (150)	0.60	0.36	0.04	94%
Nitrogen fixation (155)	0.59	0.34	0.05	78%
Fumarate reduction (88)	0.48	0.35	0.03	100%
H_2 oxidation (165)	0.47	0.27	0.05	100%
Acetogenesis (33)	0.37	0.12	0.01	94%
Aerobic chemoheterotrophy (453)	0.35	0.29	0.13	99%
Fe(III) reduction (88)	0.32	0.16	0.03	99%
DNRA (60)	0.30	0.26	0.02	97%
Mn(IV) reduction (33)	0.28	0.16	0.01	100%
Fe(II) oxidation (44)	0.28	0.13	0.01	98%
Denitrification (98)	0.22	0.07	0.03	96%
Methylotrophy (51)	0.17	0.12	0.02	100%
Arsenate reduction (30)	0.15	0.05	0.01	100%

Table S16: Completeness of KEGG modules in genomes with and without each capability. Mean completeness in the 358 labeled training genomes.

Module	Category (positive genomes)	Positives	Negatives
Methanogenesis, CO_2 to methane (M00567)	Methanogen/acetogen (75 of 358)	0.49	0.06
Methanogenesis, acetate to methane (M00357)	Methanogen/acetogen (75 of 358)	0.55	0.23
Wood-Ljungdahl pathway (M00377)	Methanogen/acetogen (75 of 358)	0.48	0.30
Dissimilatory sulfate reduction (M00596)	Sulfate reduction (63 of 358)	0.81	0.15
Denitrification (M00529)	Nitrate reduction (102 of 358)	0.46	0.11
Nitrate reduction to ammonium (DNRA) (M00530)	Nitrate reduction (102 of 358)	0.47	0.12
Nitrification, ammonia to nitrite (M00528)	Ammonia oxidation (32 of 358)	0.52	0.07
Comammox (M00804)	Nitrite oxidation (12 of 358)	0.25	0.15
SOX thiosulfate oxidation (M00595)	Sulfur oxidation (131 of 358)	0.60	0.09

Table S17: Agreement between the phenotype collection and the larger dataset. Cohen's kappa for the 179 species in both sources. [a]Fewer than ten positives in one source, so not in the means. [b]159 species with an oxygen label in both sources.

Category	Positives, collection	Positives, larger dataset	Kappa, sources	Kappa, RedoxKO vs larger dataset
Sulfate reduction	27	13	0.61	0.77
Sulfur reduction	26	20	0.60	0.39
Methanogen/acetogen	60	52	0.82	0.85
Fe(III) reduction[a]	21	7	0.47	0.14
Nitrate reduction	51	44	0.73	0.55
H_2 oxidation	125	33	0.14	0.11
Sulfur oxidation	67	44	0.68	0.59
Fe(II) oxidation	21	11	0.59	0.20
O_2 respiration[b]	72	75	0.73	0.87
Mean			0.61	0.54
Mean without H_2 oxidation			0.68	0.60

Table S18: LLMs and trained models on the same 96 genomes. Input gives what each model received. Bold marks the best value for each task. [a]84 genomes, because the run stopped when the API budget ran out.

Model	Input	Macro AUPRC
Donor and acceptor categories (14)		
Taxonomic baseline	taxonomy	0.485
Claude	KOs	0.578
GPT-6 Astra	KOs	0.588
RedoxKO	KOs	0.680
RedoxKO-ens	KOs	0.687
RedoxKO-ens-BF	KOs, Bacformer	0.697
Claude	KOs + organism name	**0.777**
GPT-6 Astra[a]	KOs + organism name	0.731
Pathway traits (20)		
Taxonomic baseline	taxonomy	0.531
Claude	KOs	0.471
Claude	KOs + organism name	0.663
EnergyKO-M	KOs + modules	**0.675**

Table S19: Released models and their uses. Times are per genome from an NCBI accession on a CPU unless stated. File gives the name in the release.

Model	Predicts	Input and learner	Output	Use when	File
RedoxKO	14 donor and acceptor categories	852-KO panel, random forest, about 15 s	Score, calls in 6 categories, confidence label	Default, ranking genomes for testing, calls when false positives are costly	RF
RedoxKO-LX	Same	852-KO panel, logistic regression or XGBoost per category	Score, calls	First final model, for comparison	B2
RedoxKO-ens-BF	Same	KO panel and Bacformer-large vector, protein embeddings on a GPU, about 41 s	Score, calls	Bacformer-large vectors already available	B2XLE_large
EnergyKO-M	20 pathway traits	Full KO profile and KEGG module completeness, logistic regression or XGBoost per trait, about 95 s	Score, calls in 11 traits[a]	Pathway traits	PB3
EnergyKO-BF	Same	EnergyKO and Bacformer-large vector	Score	Bacformer-large vectors already available	PE

[a]Acetogenesis, fermentation, hydrogen oxidation, iron oxidation, nitrate reduction, nitrogen fixation, photoautotrophy, photoheterotrophy, sulfate reduction, sulfur oxidation and sulfur reduction.

Table S20: Blind tests of the frozen final model. Species from description papers, scored once after the model was frozen on 22 October 2026. Macro AUROC is averaged over categories with positive and negative tests. MCC is for yes/no calls in the six categories that get them.

Test set	Species	Tests (negative)	Macro AUROC, RedoxKO	Macro AUROC, RedoxKO-LX	MCC, RedoxKO	MCC, RedoxKO-LX
Species described in 2026, oxygen respiration only	49	51 (6)	0.97[a]	0.89[a]	0.56	0.36
Lithotrophic species, all categories	76	168 (38)	0.76	0.74	0.41	0.46
Lithotrophic species, electron donors, both tiers	47	52 (7)	0.61	0.58	—	—

Test set	Species	Tests (negative)	Macro AUROC, RedoxKO	Macro AUROC, RedoxKO-LX	MCC, RedoxKO	MCC, RedoxKO-LX
Electron donors, second tier only[b]	44	49 (6)	0.79	0.81	—	—

[a]Oxygen respiration only, the one category with positive and negative tests. [b]Genera absent from the training genomes but present in other project data. Of the 88 lithotrophic species, 76 kept at least one test.

Table S21: Paired comparisons and controls for the donor and acceptor models. Differences in macro AUPRC, or in MCC where stated. Positives backed by growth experiments are those traced to statements graded A or B (506 of 1,016 positives).

Comparison	Data	Value	95% CI
RedoxKO minus the nearest-neighbor rule			
RedoxKO minus the rule	Genus-level CV	+0.040	[−0.0004, +0.075]
RedoxKO minus the rule	Family-level CV	+0.030	[−0.003, +0.078]
RedoxKO minus the rule	External genomes	+0.119	[0.080, 0.154]
Positives backed by growth experiments only			
RedoxKO	Genus-level CV	0.517	
RedoxKO	Family-level CV	0.477	
Taxonomic baseline	Genus-level CV	0.380	
Taxonomic baseline	Family-level CV	0.290	
RedoxKO minus taxonomic baseline	Genus-level CV	+0.136	[0.096, 0.190]
RedoxKO minus taxonomic baseline	Family-level CV	+0.187	[0.138, 0.240]
Same difference with all positives	Genus-level CV	+0.140	
Same difference with all positives	Family-level CV	+0.182	
Same difference with all positives	External genomes	+0.174	[0.138, 0.218]
Model against model, folds of this paper			
RedoxKO minus RedoxKO-LX	Genus-level CV	+0.037	[−0.001, +0.059]
RedoxKO minus RedoxKO-LX	Family-level CV	+0.002	[−0.039, +0.040]
RedoxKO minus RedoxKO-LX	External genomes	+0.043	[0.018, 0.069]
RedoxKO-ens-BF minus RedoxKO-ens	Genus-level CV	+0.023	[−0.003, +0.047]
RedoxKO-ens-BF minus RedoxKO-ens	Family-level CV	+0.012	[−0.020, +0.027]
RedoxKO-ens minus RedoxKO-LX	External genomes	−0.007	[−0.022, +0.009]

Comparison	Data	Value	95% CI
RedoxKO-ens minus RedoxKO-LX	External, labels not from FAPROTAX	−0.050	[−0.080, −0.014]
Bacformer-large minus RedoxKO-LX	External genomes	−0.071	[−0.109, −0.029]
RedoxKO minus RedoxKO-LX, MCC	100 new species	+0.056	[0.006, 0.115]
METABOLIC minus RedoxKO, MCC	100 new species	+0.10	[−0.02, +0.22]
METABOLIC minus RedoxKO, MCC	43 withheld species	+0.07	[−0.11, +0.23]
METABOLIC minus RedoxKO, MCC	100 new species, six call categories	+0.04	[−0.10, +0.18]
DRAM minus RedoxKO, MCC	100 new species, six call categories	+0.06	[−0.06, +0.18]
Significant differences over the six fold allocations			
RedoxKO minus RedoxKO-LX	Genus-level CV	1 of 6	
RedoxKO minus RedoxKO-LX	Family-level CV	0 of 6	
RedoxKO-ens-BF minus RedoxKO-ens	Genus-level CV	0 of 6	
RedoxKO-ens-BF minus RedoxKO-ens	Family-level CV	0 of 6	

Supplementary figures

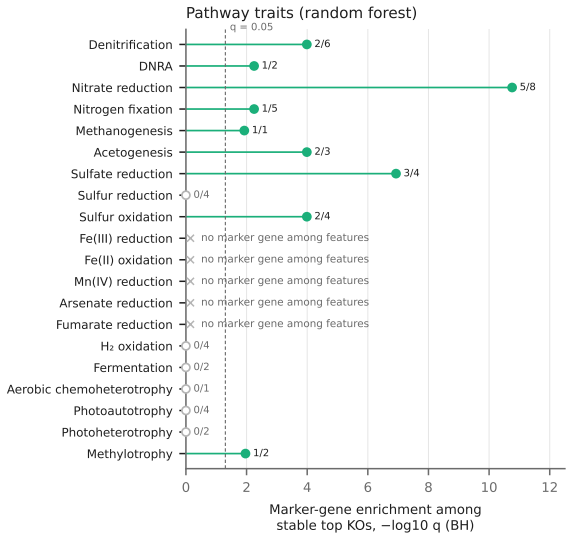


Figure S1: Top features of the pathway models and their agreement with known markers. Enrichment of marker genes among the most stable features of random forests for the 20 pathway traits. Grey points mark traits without significant enrichment, and crosses traits without a marker among the features. Labels such as 2/6 count marker genes among these features.

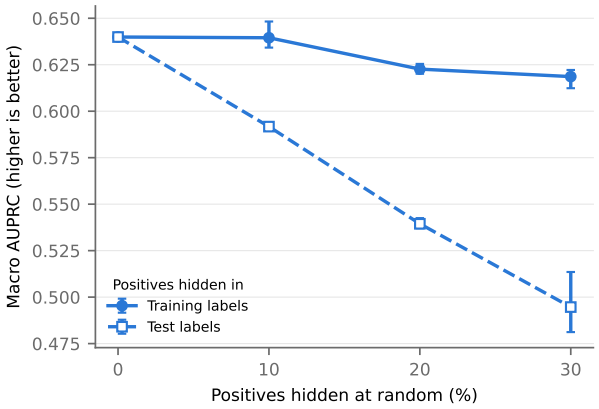


Figure S2: Effect of unreported positives. Positive labels were hidden at random in the training genomes before training or in the test genomes at scoring. Macro AUPRC of a random forest on the KO panel (300 trees, one seed) in genus-level cross-validation, averaged over folds within each category, so higher than in Table 1. Error bars show the range of three repeats.